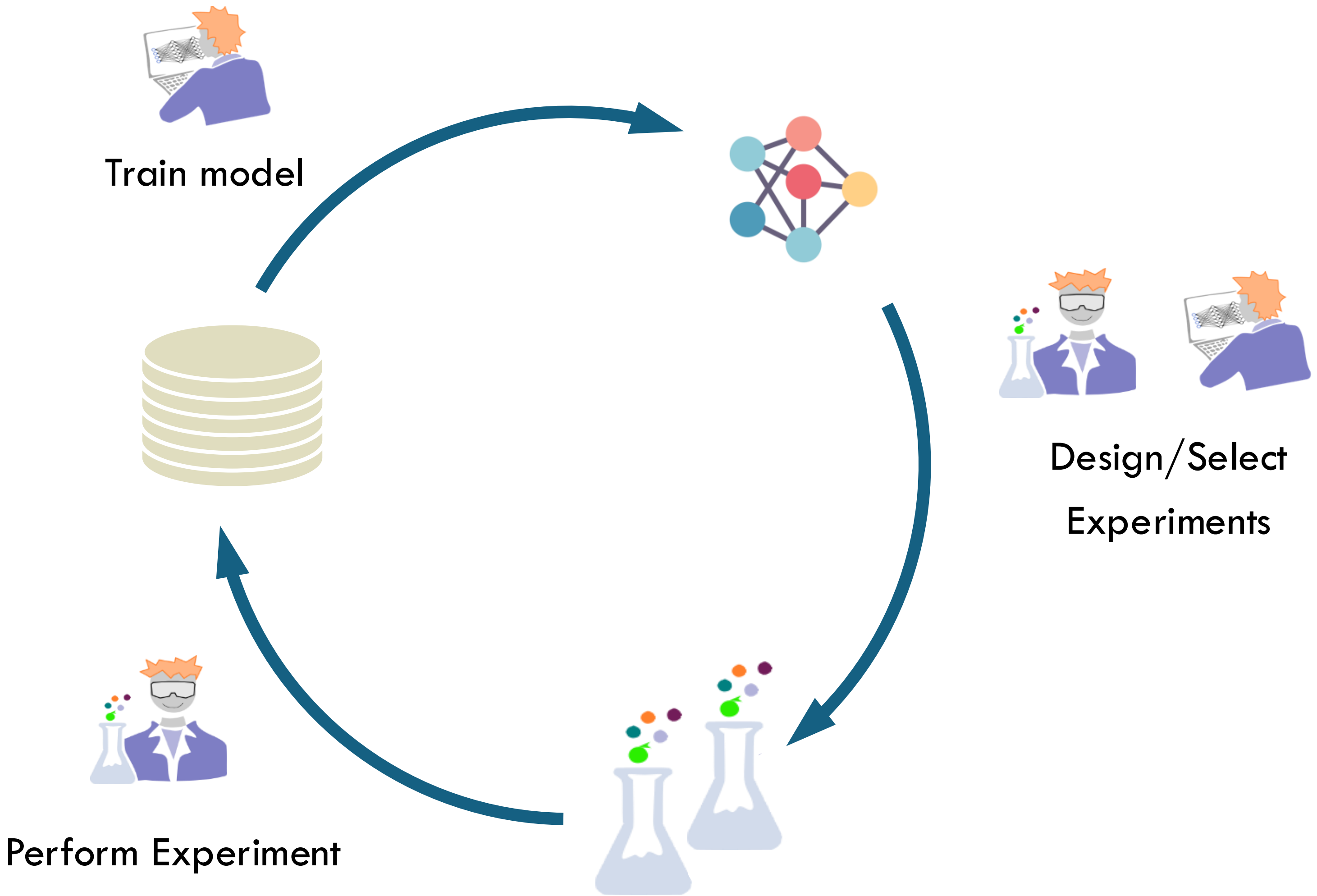
Active Learning



J. Schleinitz
Some slides were created using Claude Code

Plan

1 · The BO/AL loop

2 · Objective = model improvement

3 · What changes in the acquisition function

4 · A worked example: regioselectivity

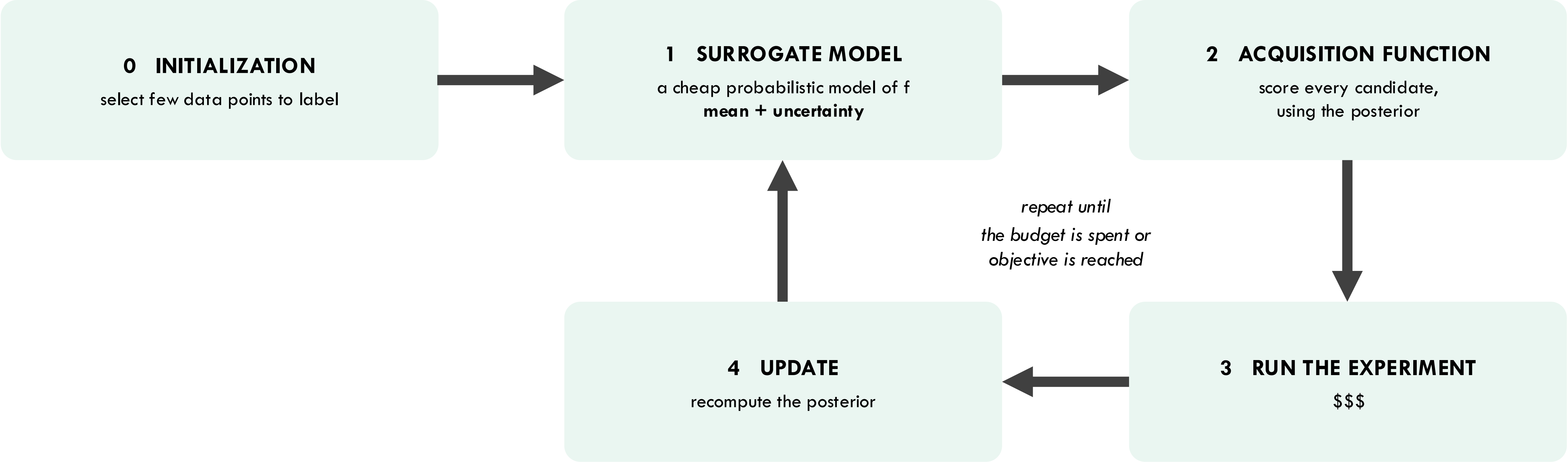
5 · Which uncertainty is worth buying?

6 · Practical issues

7 · Active learning in the wild — and what is not

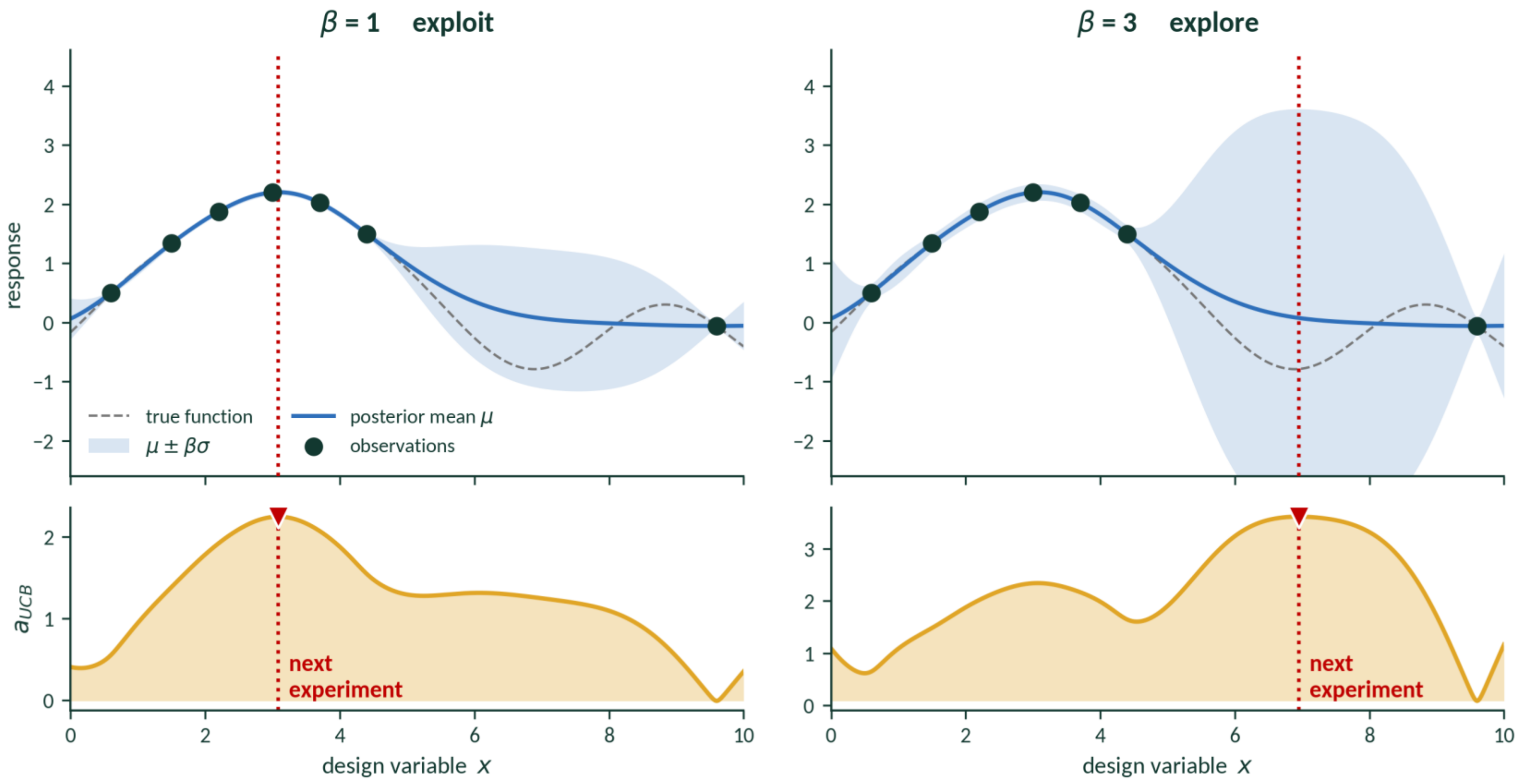
8 · Conclusion

The loop


0 INITIALIZATION
select few data points to label
1 SURROGATE MODEL
a cheap probabilistic model of f
mean + uncertainty
2 ACQUISITION FUNCTION
score every candidate,
using the posterior
3 RUN THE EXPERIMENT
$$$
4 UPDATE
recompute the posterior
repeat until
the budget is spent or
objective is reached

Bayesian Optimization Acquisition Function

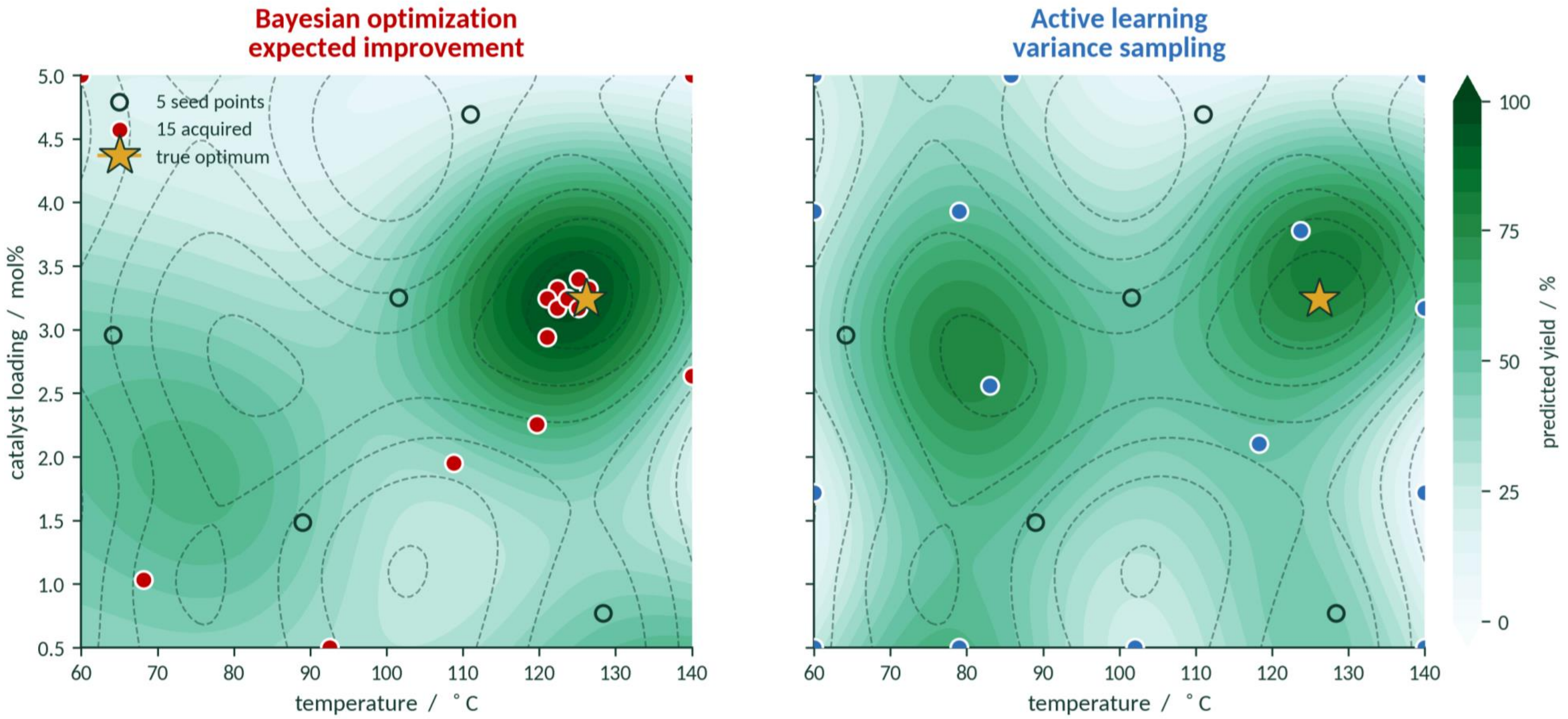
$$a_{UCB}(x) = \mu(x) + \beta\,\sigma(x)$$



β small → trust the mean, stay near the best point. **β large → buy the error bar, go somewhere new.**

Plan

BO and AL: same loop but different question



best yield found 95%
RMSE away from optimum 15.9%
better optimum in 28/30 runs

best yield found 79%
RMSE away from optimum 9.5%
better model away from it in 28/30 runs

When the model is the deliverable

You are going to interpret the model.

Interpretability. If you will pull SHAP values or a rate law out of it, it has to be right in the whole region you will interpret.

You are going to run it a million times.

Deployment at scale. An interatomic potential, a regioselectivity model, a pKa predictor. Its value is accuracy across the whole domain, on inputs nobody has looked at.

Plan

Acquisition Function depends on Model Objective

you are predicting

A value, across a library

→ **uncertainty sampling**

$\sigma(x)$ — GP variance or ensemble spread

You want low error everywhere.
Sampling where the model knows least.

you are predicting

A class

→ **margin**

$P(\hat{y}_1 | x) - P(\hat{y}_2 | x)$

You want the decision right.
Sampling where the top two are closest to a tie.

you are predicting

A ranking

→ **margin, again**

score(top site) − score(runner-up)

Which C–H reacts?
An argmax over sites — so the same criterion applies.

I. Optimizing Regressor Prediction across Chemical Space

Gaussian process

$$x^* = \text{argmax}\ \sigma^2(x)$$

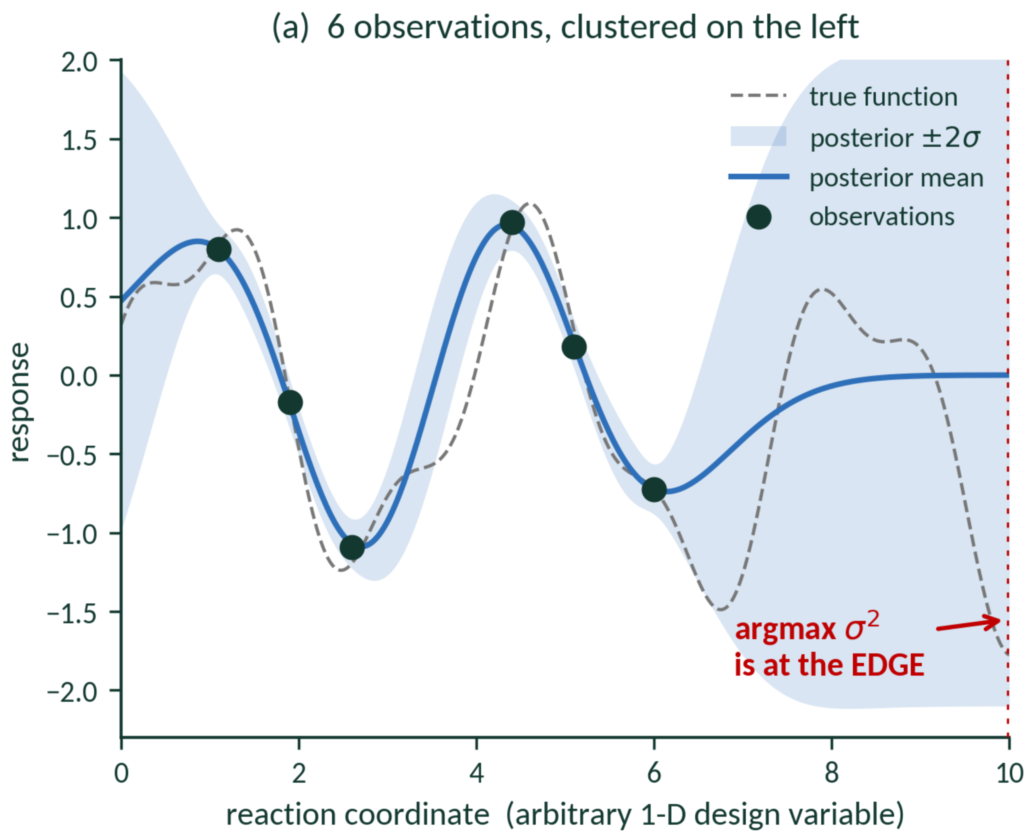
Closed form, no retraining, no label needed to evaluate it. Ideal for small data.

Any ensemble

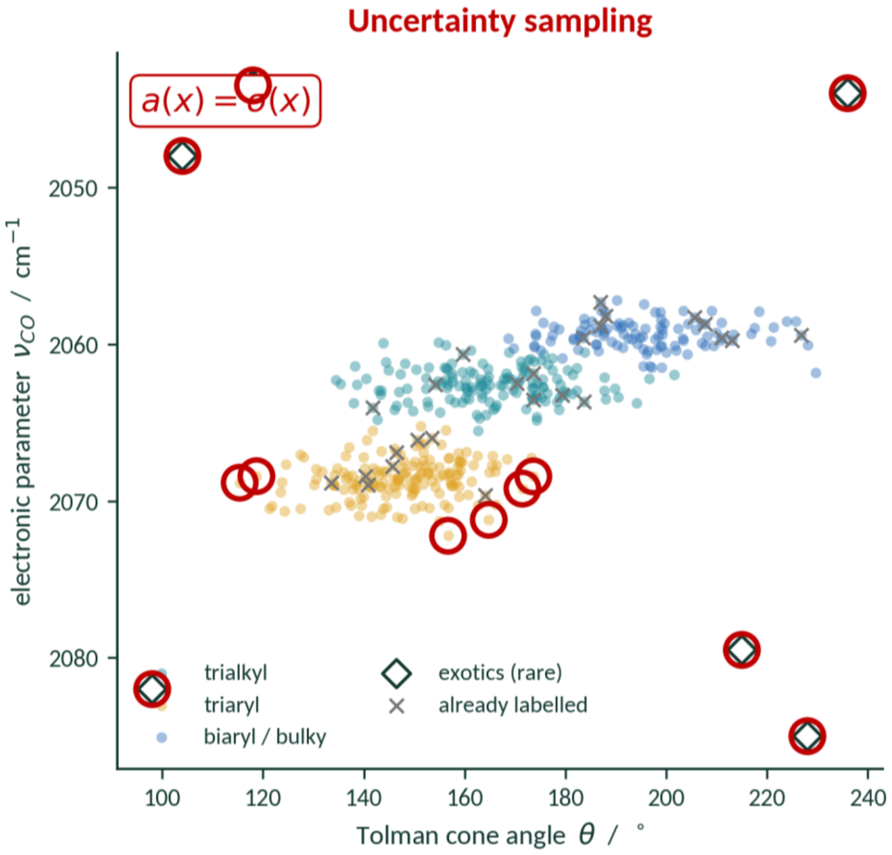
$$\sigma^2(x) = (1/M)\ \Sigma_m\ (\ f_m(x) - \bar{f}(x)\ )^2$$

Model-agnostic — random forests, D-MPNN, neural-network potentials. Costs M× training.

Uncertainty sampling reaches for outliers

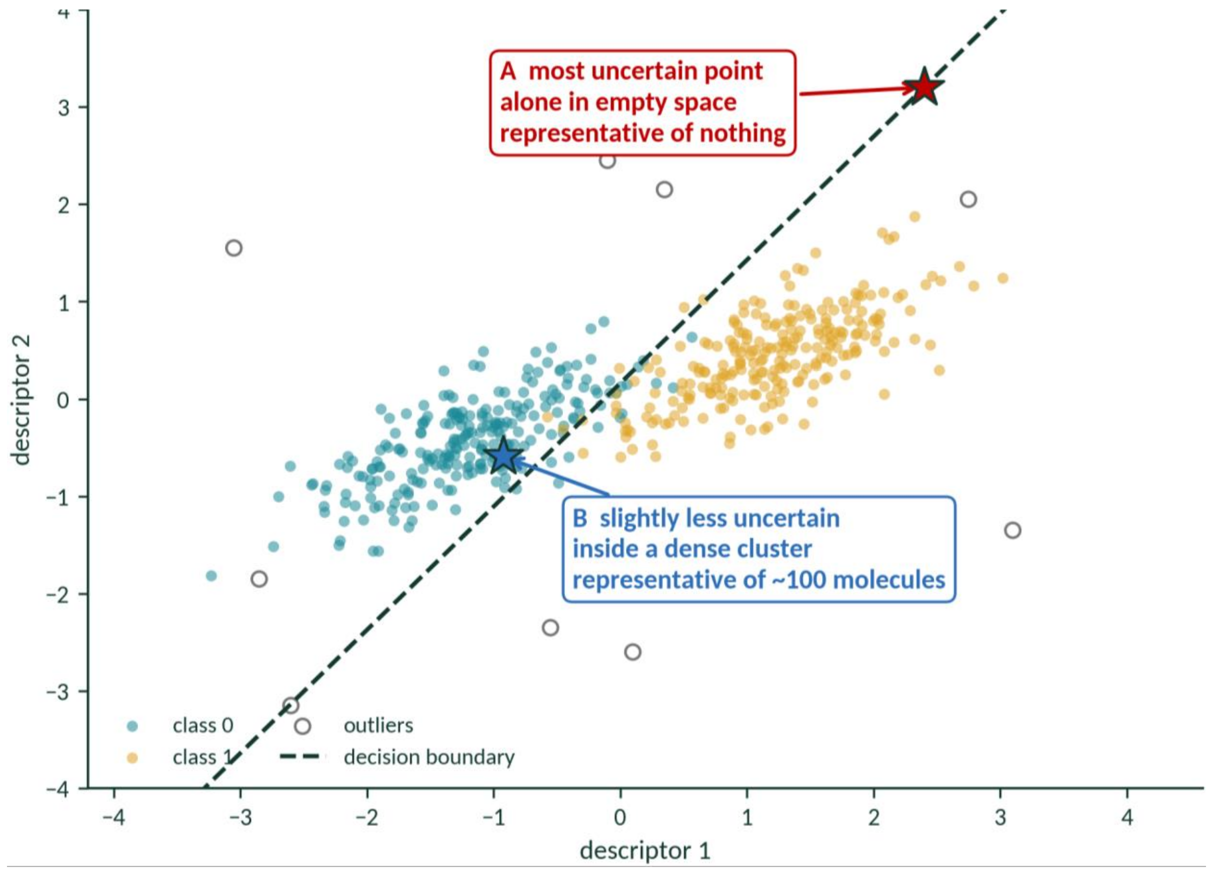


Uncertainty sampling reaches for outliers



After Settles, Active Learning Literature Survey, TR1648, 2009, Fig. 7 · information density: Settles & Craven, EMNLP 2008

Bias towards Representativeness



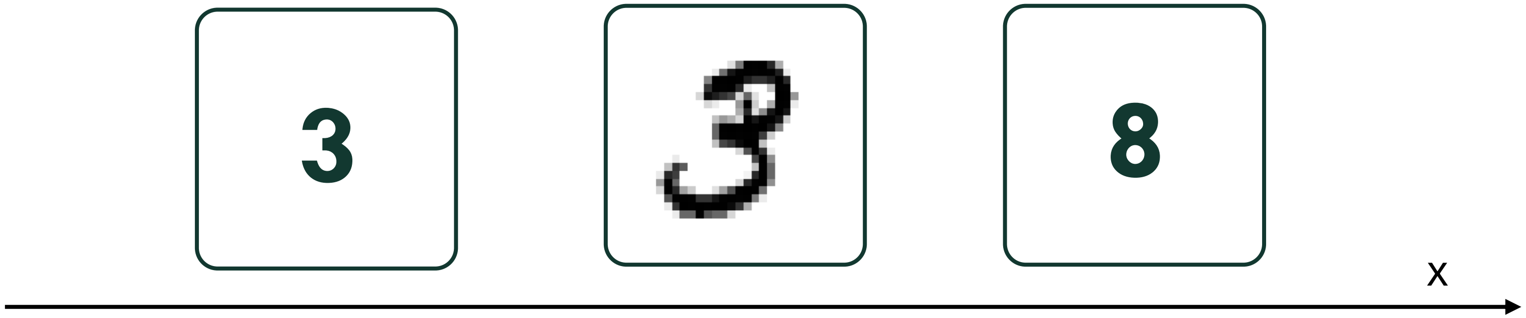
$$a(x) = \sigma(x) \cdot \text{density}(x)^b$$

After Settles, Active Learning Literature Survey, TR1648, 2009, Fig. 7 · information density: Settles & Craven, EMNLP 2008

II. Classification task

Margin

$$x^* = \text{argmin} \left[P(\hat{y}_1 \mid x) - P(\hat{y}_2 \mid x) \right]$$



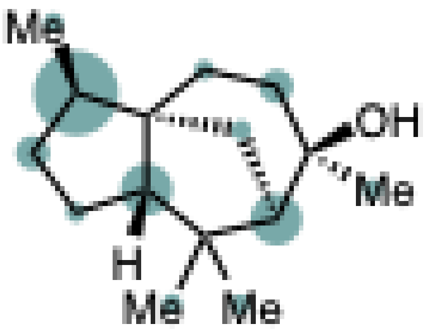
P(3|x) ~ 1

P(3|x) ~ 0.5
P(8|x) ~ 0.5

P(8|x) ~ 1

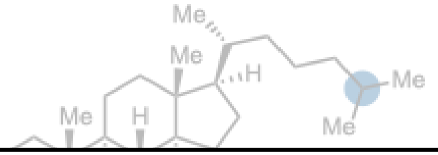
Which C–H bond reacts? An argmax over the sites in one molecule.

margin = score(best site) − score(runner-up)

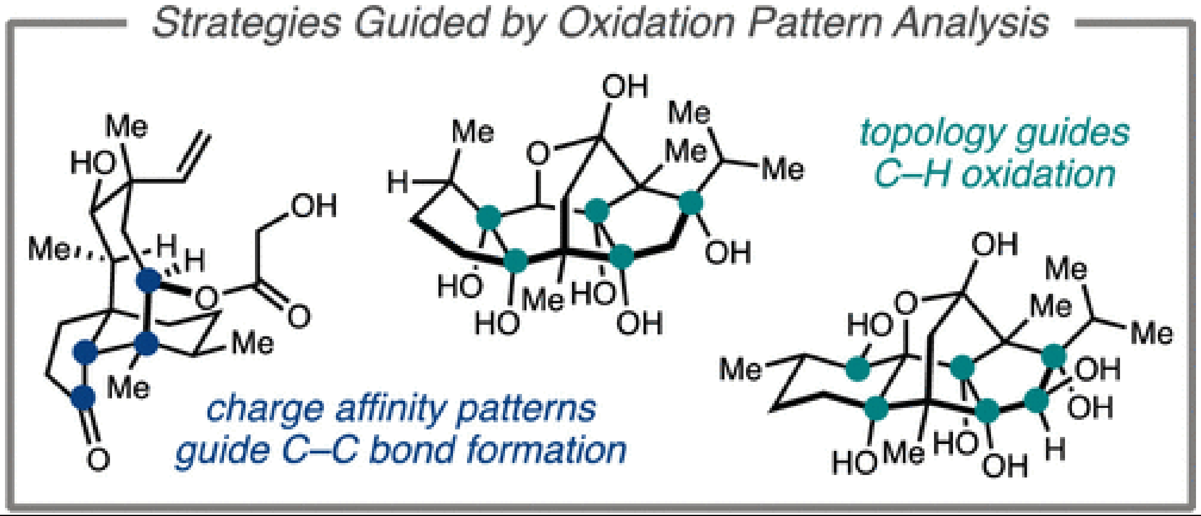


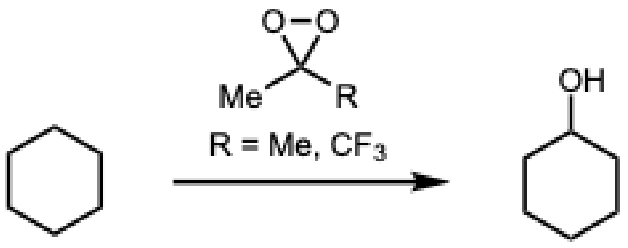
Plan

LSF: introduction of functionality



site selectivity

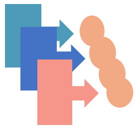




Dioxirane oxidations

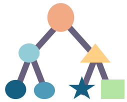
Predicting site-selectivity: previous work

Transformer



Neural Network

Decision Tree



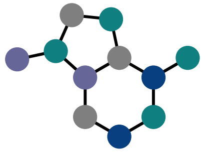
Linear Regression

Decreasing model complexity

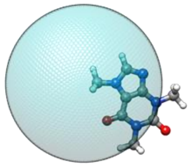
Increasing representation complexity

C2=CC=C(Br)C=C2

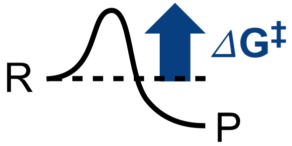
SMILES



Graphs



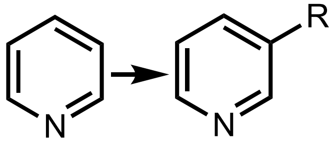
xTB



Transition States

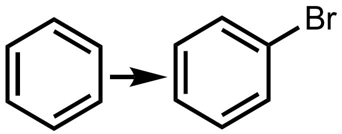
General reaction outcome prediction

Accuracy: ~83% for brominations



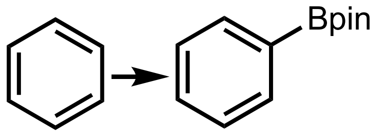
Arene borylation

Accuracy:
~90% in sample
~70% out of sample



EAS

Accuracy:
~93% in sample
~90% out of sample

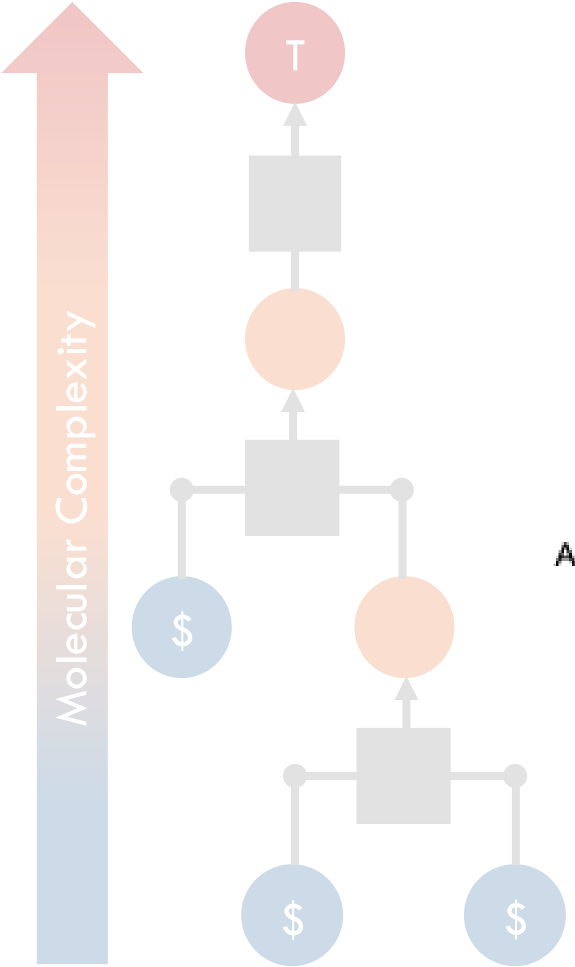


Ir-cat. arene borylation

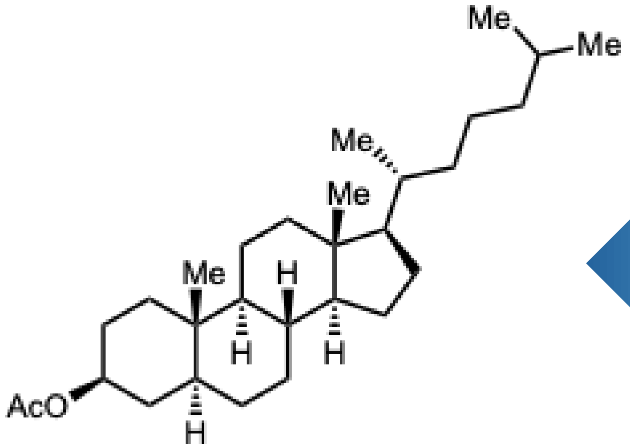
Accuracy: ~92%

Ree, N. *et al. Digital Discovery*, **2022**, *1*, 108. Nippa, D. F. *et al. Chemrxiv*, **2023**, Nippa, D. F. *et al.*, *Chemrxiv*, **2022**. Caldeweyher, E. *et al.*, *JACS*, **2023**, *145*, 17367–17376. Schwaller P., *et al.*, *ACS Cent. Sci.*, **2019**, *5*, 1572.

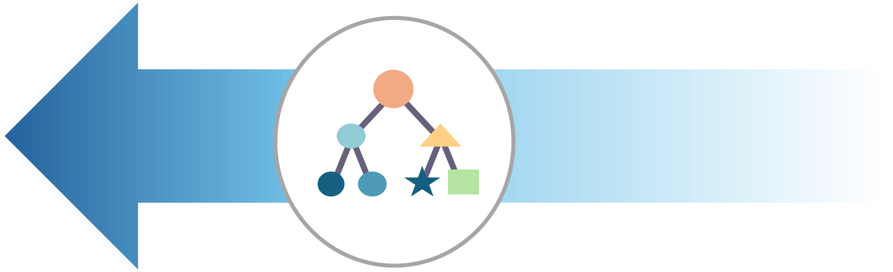
Predicting site-selectivity on complex molecules



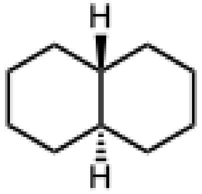
Complex molecules

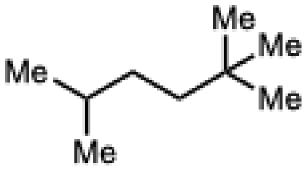


- difficult to access
- hard to characterize
- prediction: high value



Small molecules

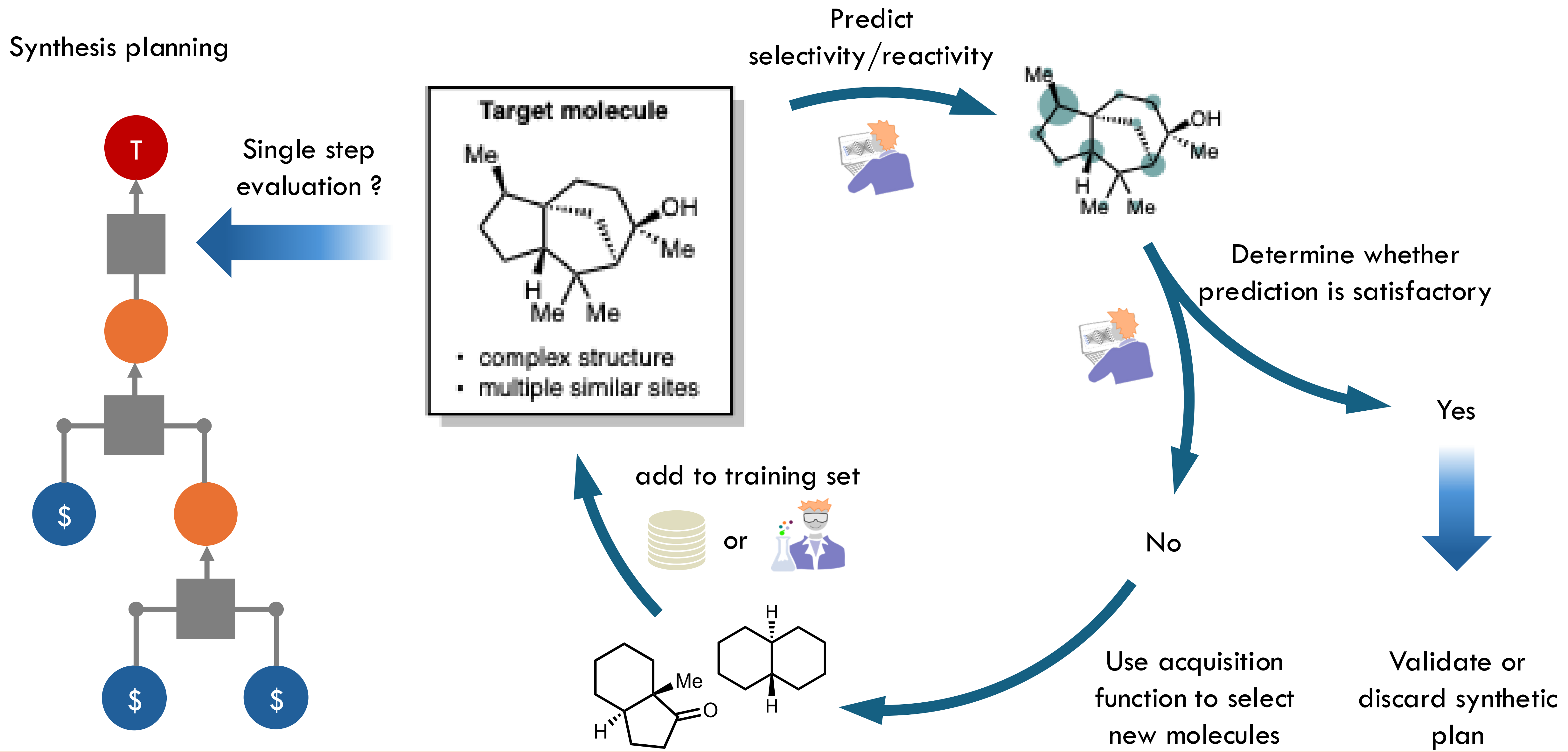


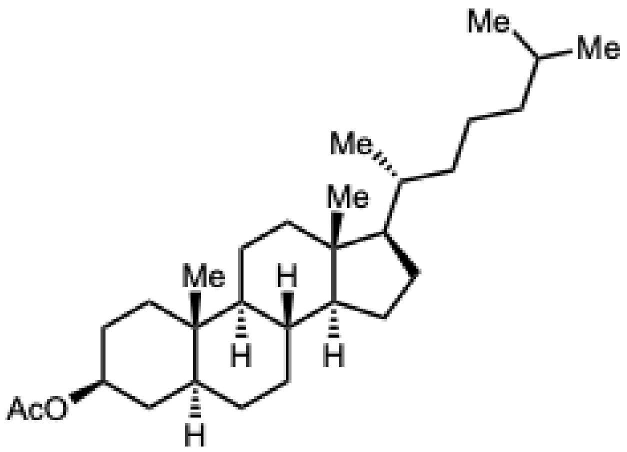


- commercially available
- easy to characterize
- prediction: low value

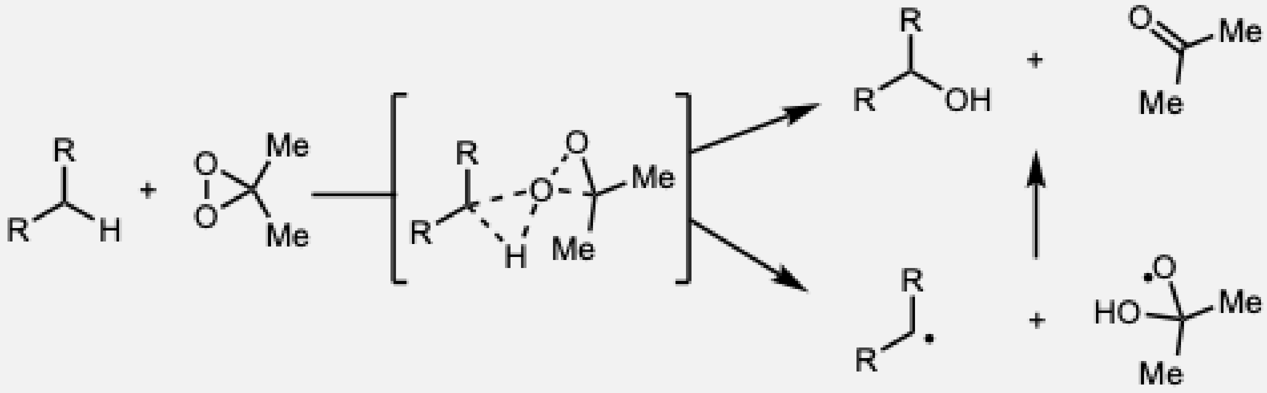
How can we be sure that a model will predict well on a target of interest?

Perspective for de-risking synthetic planning

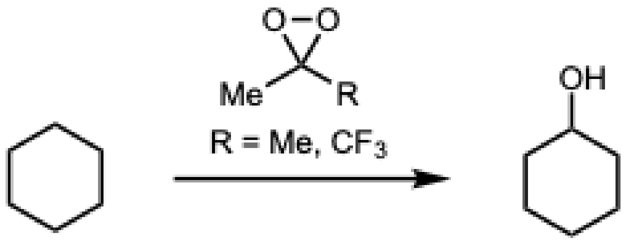




Calculate transition states for all C–H bonds?

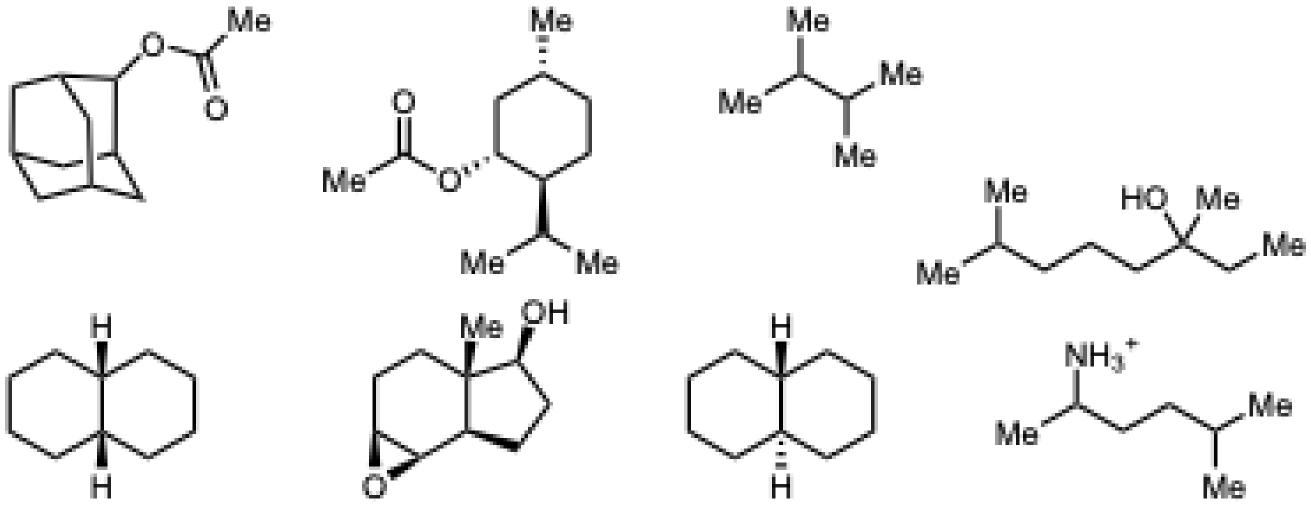


C–H bond oxidation database

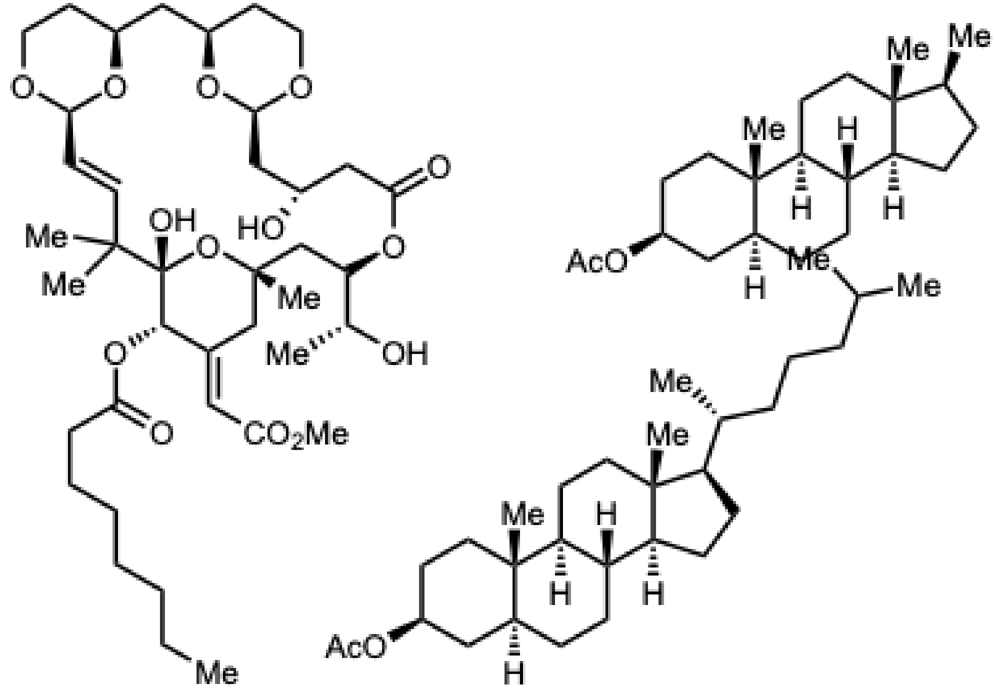


Literature database - dioxirane oxidations
~130 small molecules
~50 complex molecules

Small molecules < 15 C

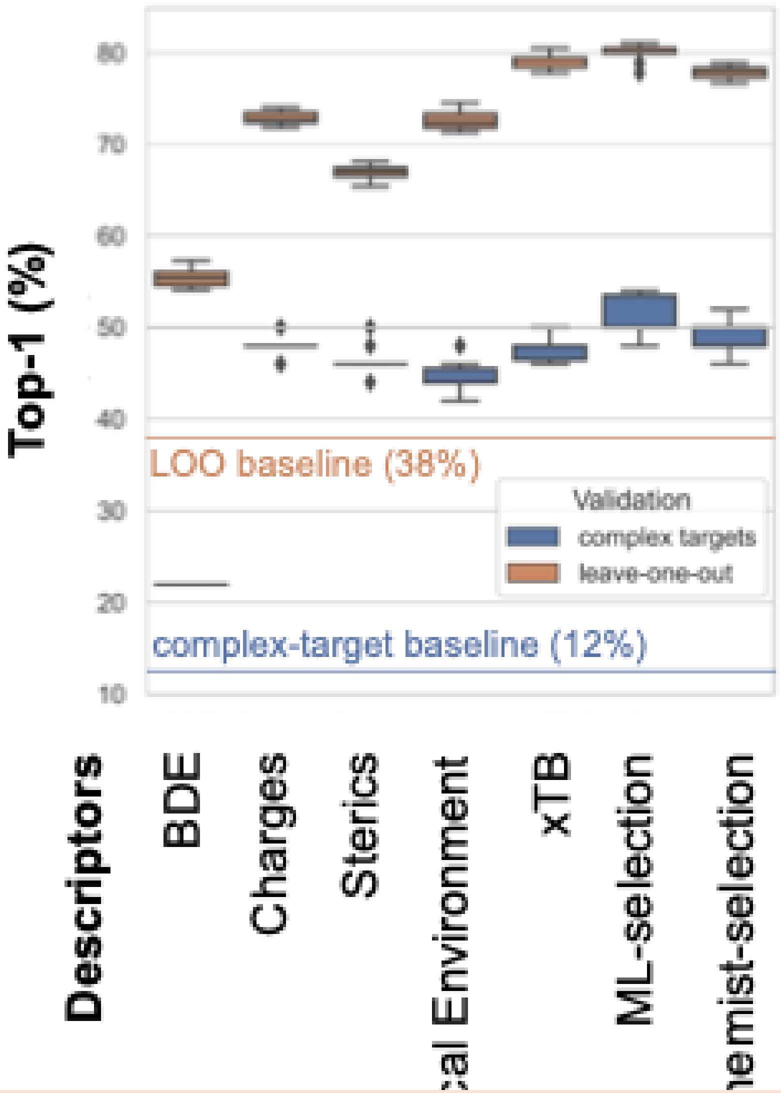


Complex molecules >= 15 C



Predicting site-selectivity validation?

Top-1 model performance



- Textbook baseline:

Benzylic > tertiary > secondary > primary

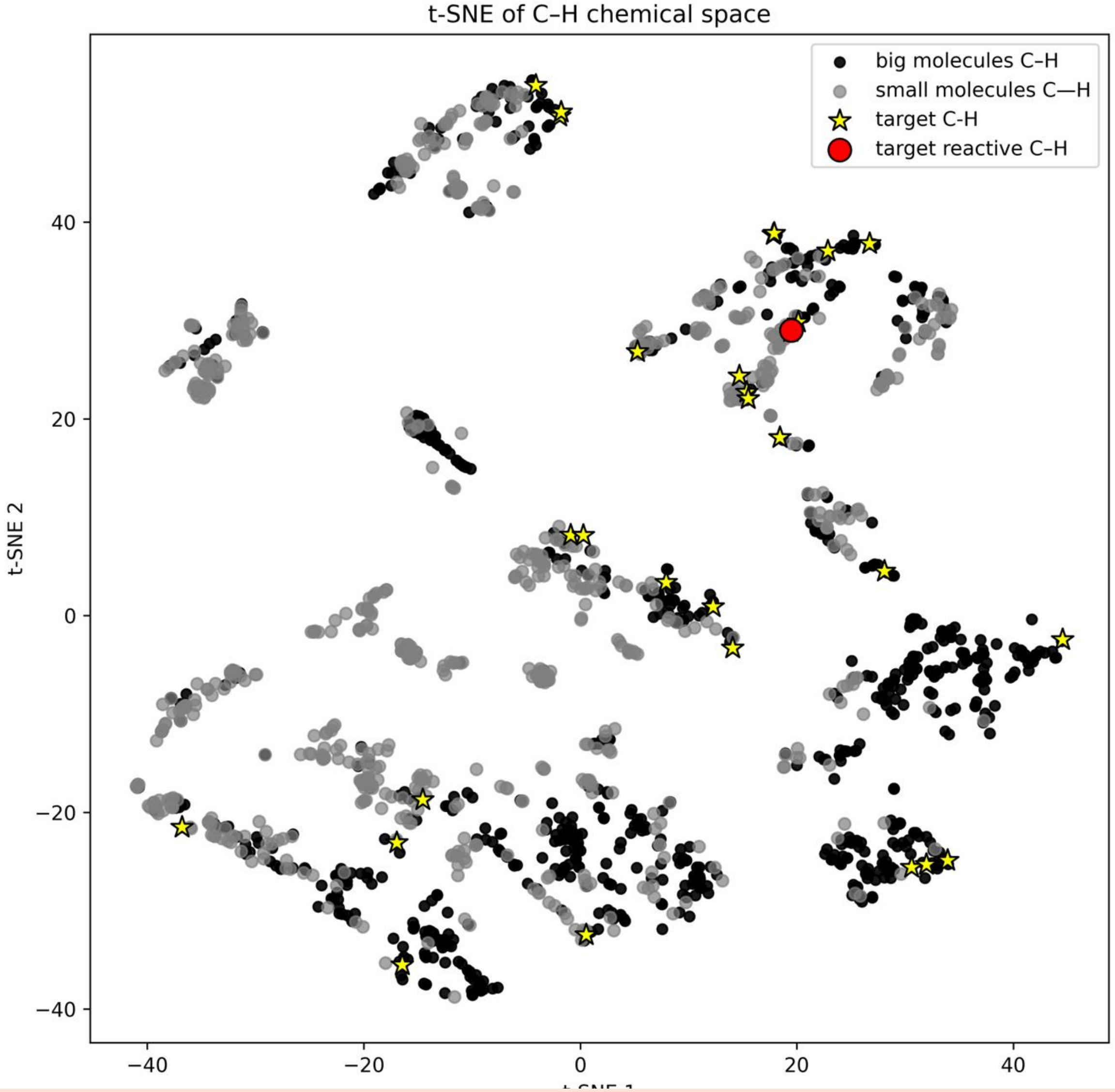
Validations:

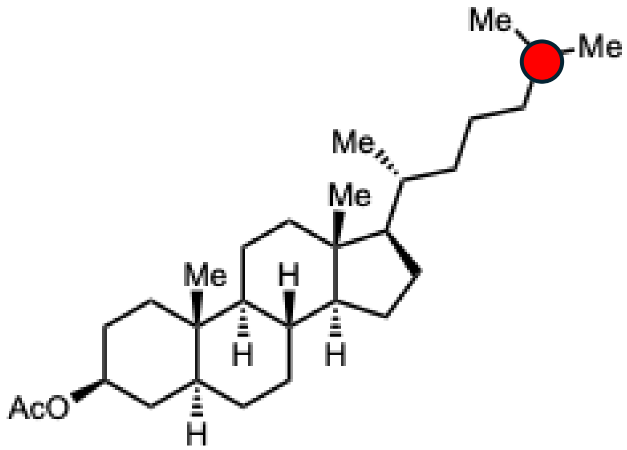
- **Leave one out (LOO)** for 185 molecules.
- **Complex-target validation:**

Train on small molecules (135 molecules) and predict on complex molecules (50 molecules)

C–H chemical space

Stereo-electronic representation:
- BDE
- C-H charges
- %Vbur



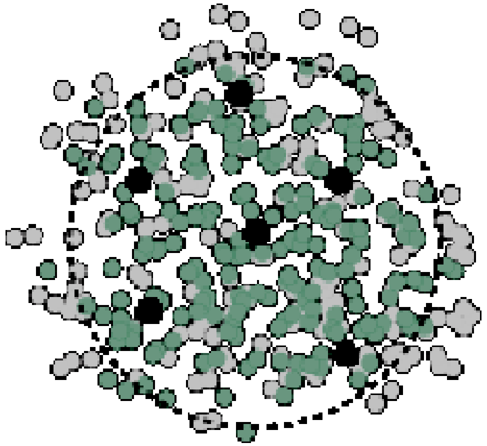


Target molecule

Predicting site-selectivity on complex molecules

1. Large foundational model

- Data expensive
- Modeling Challenges



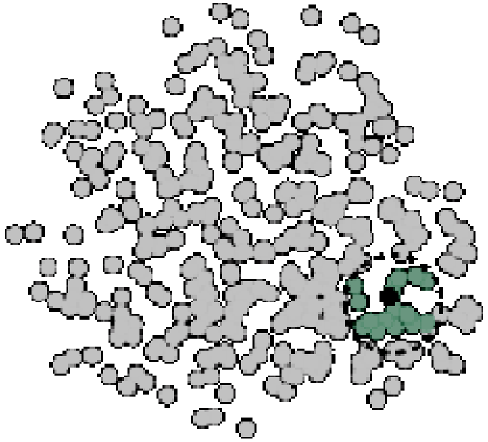
Applicability domain

Training data

Targets

2. Target oriented dataset design:

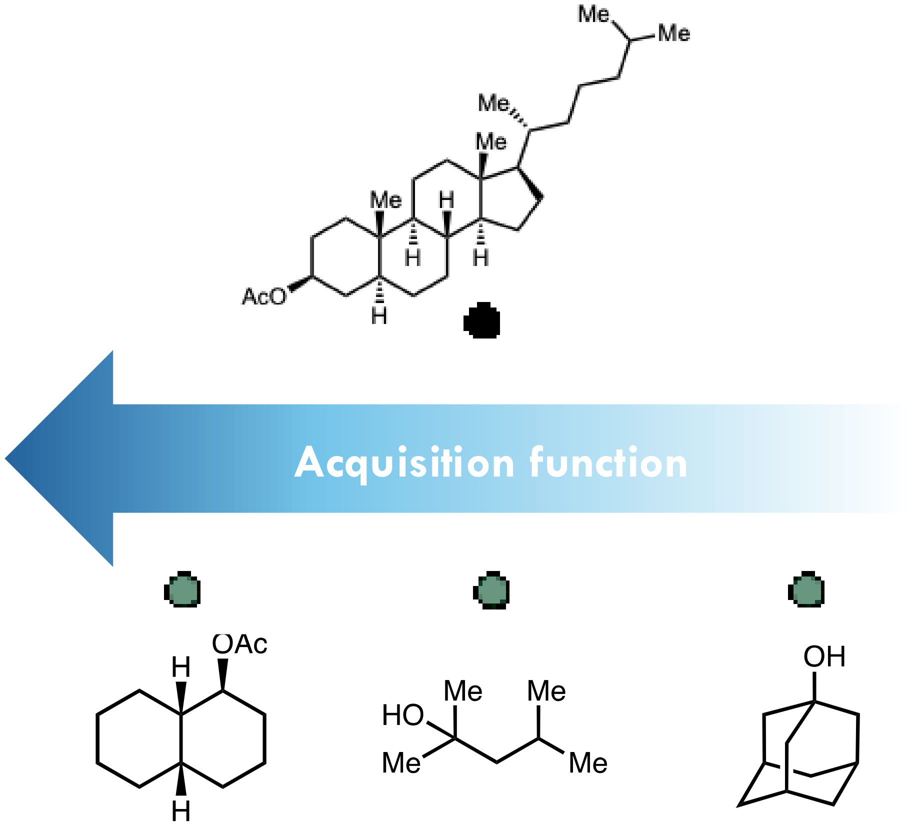
- Small training set representative of the target reactivity



Applicability domain

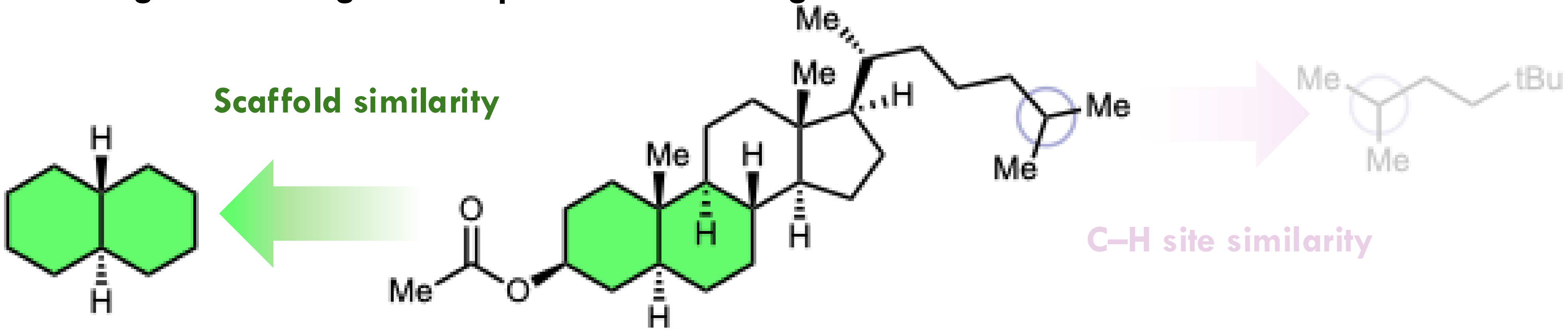
Training data

Target

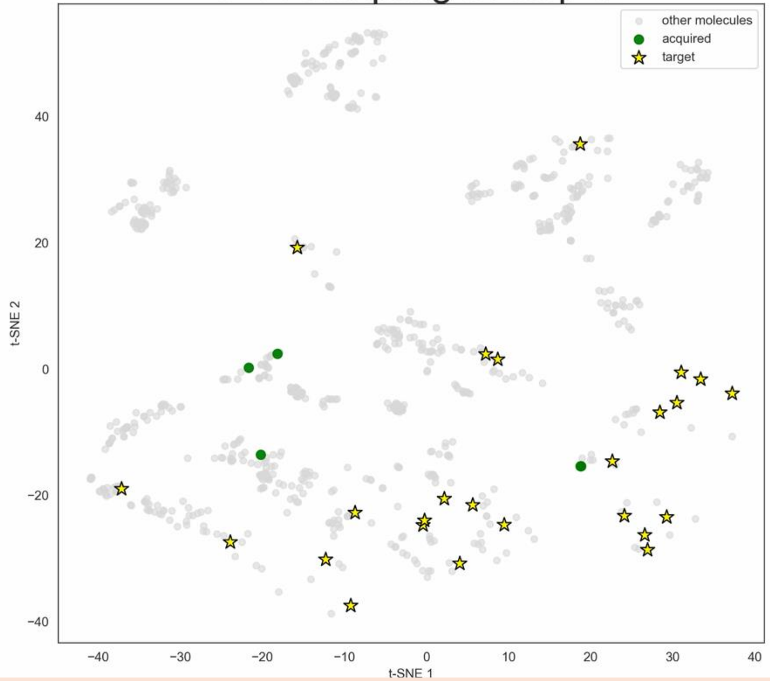


Active learning baselines: similarity

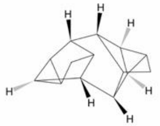
What makes a training molecule "good" for prediction on a target?



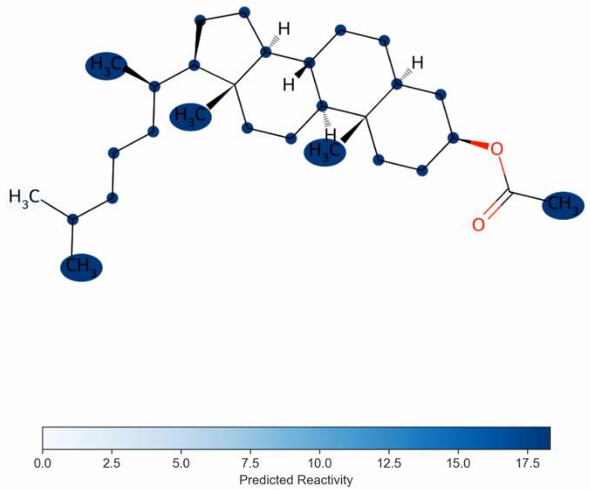
C–H sampling at step 1



Tested molecules at step 1



Model prediction at step 1



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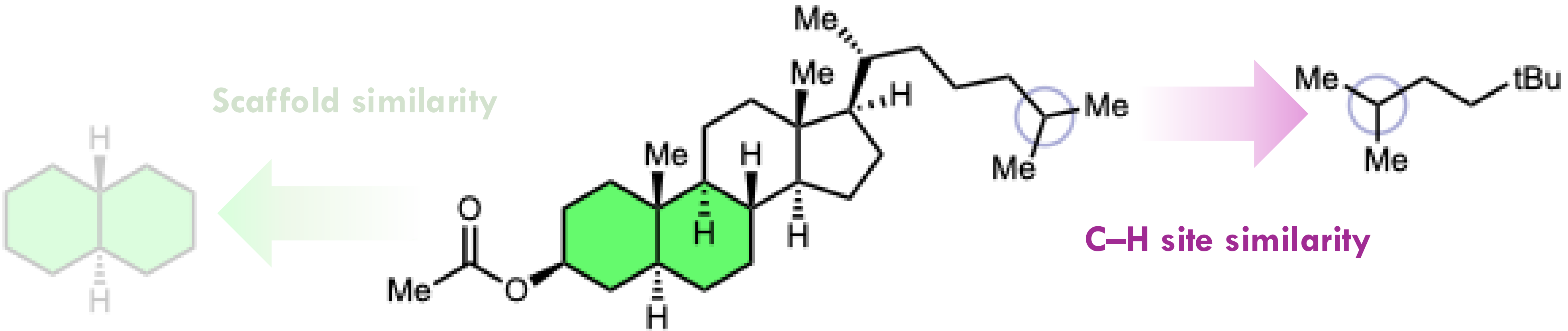
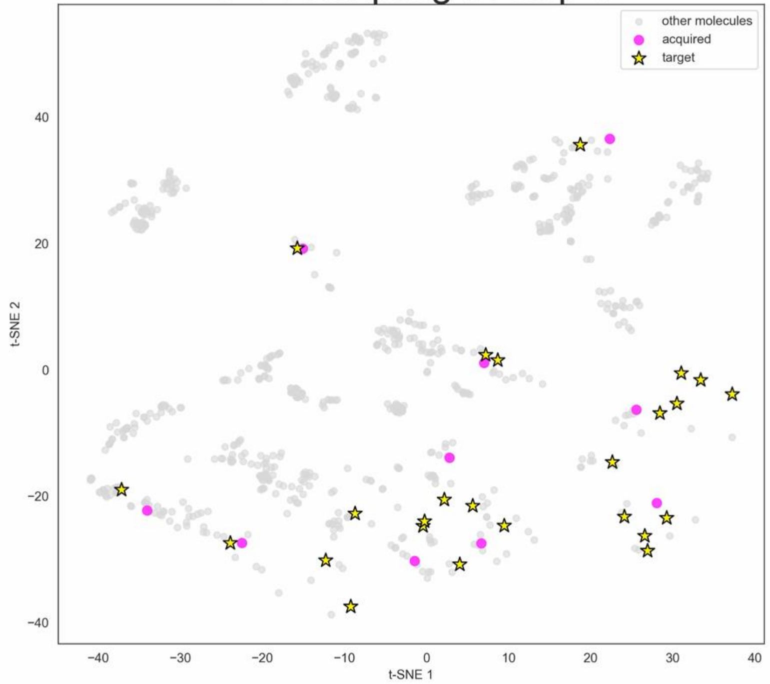
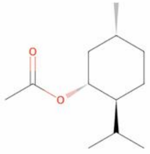


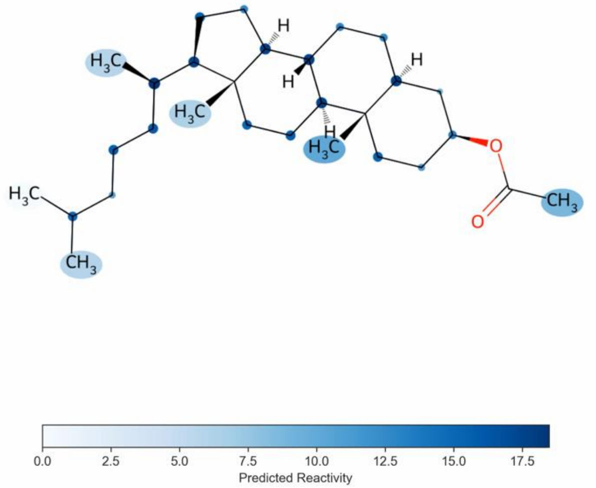
C–H sampling at step 1



Tested molecules at step 1



Model prediction at step 1



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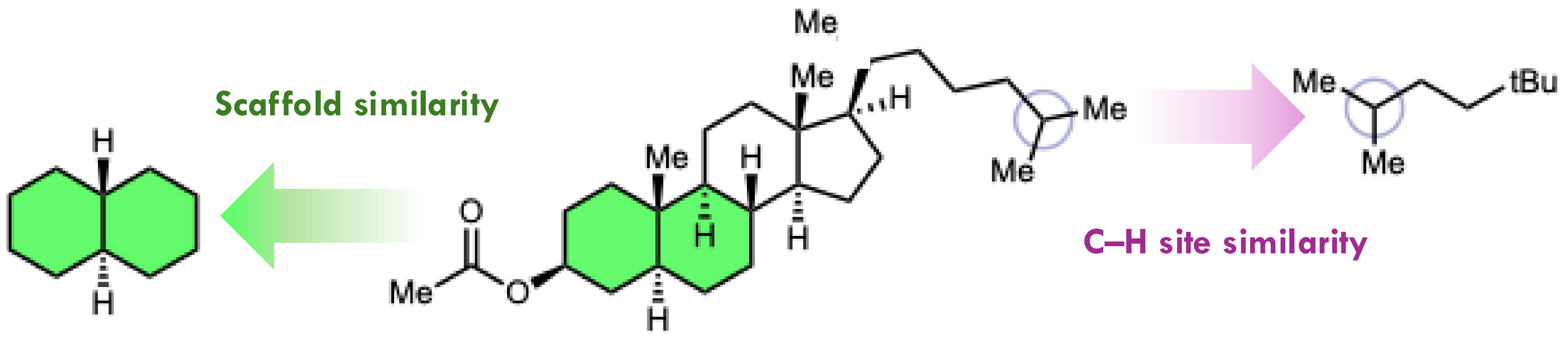
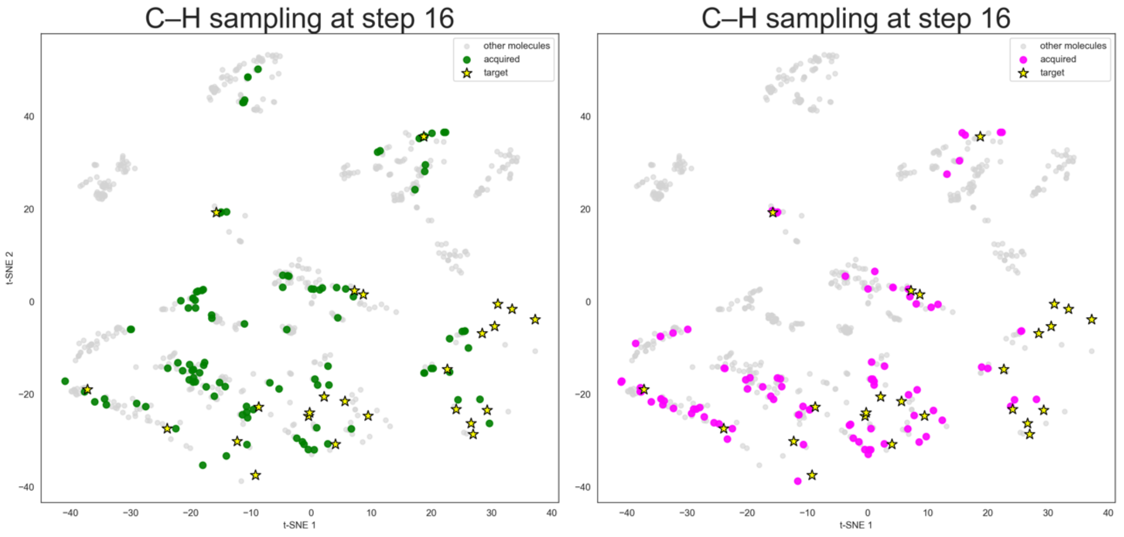
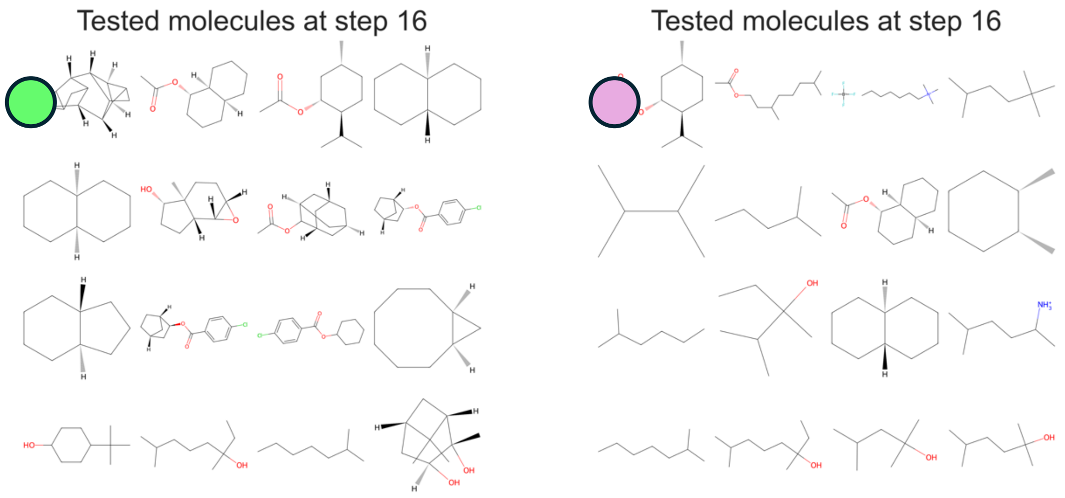




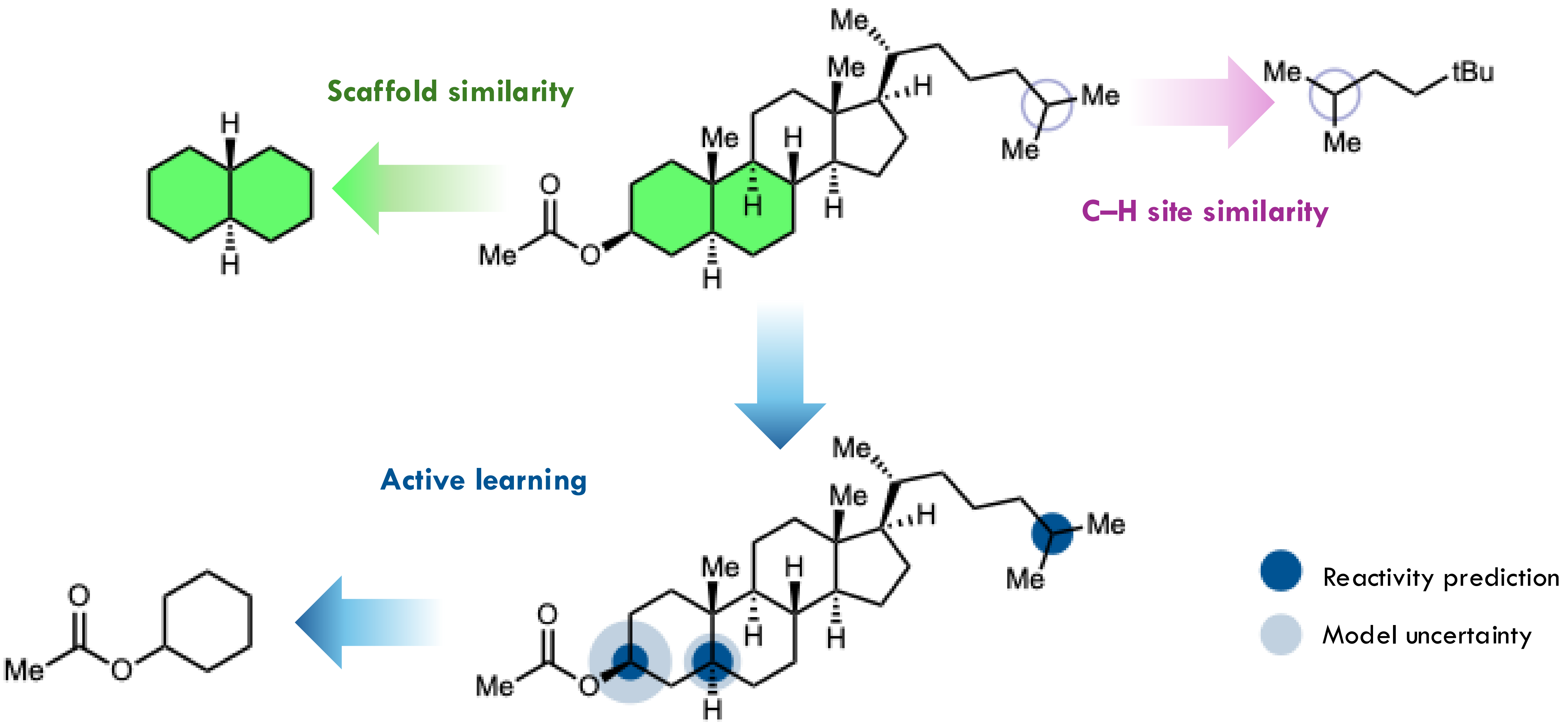


Pro: Easy to implement, very intuitive

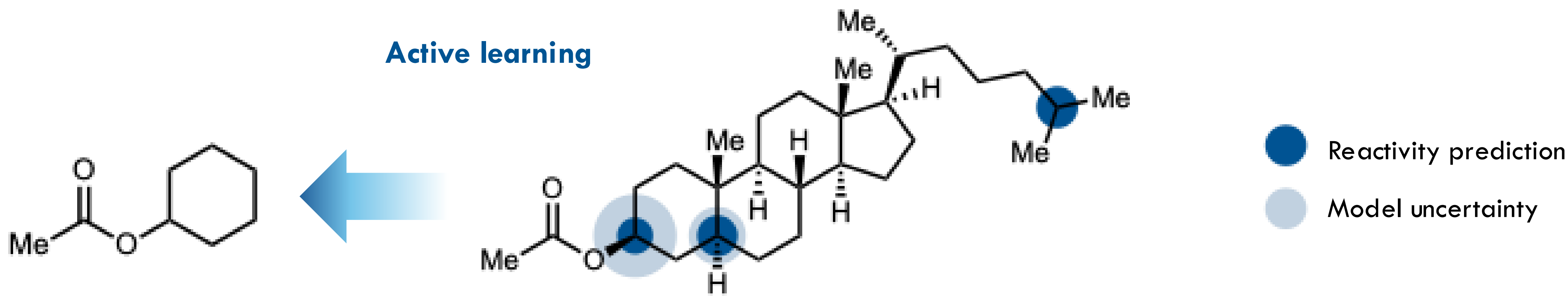
Cons: Redundancy of the information, not taking into of what the model learned previously

Active learning: leveraging uncertainty and predictions

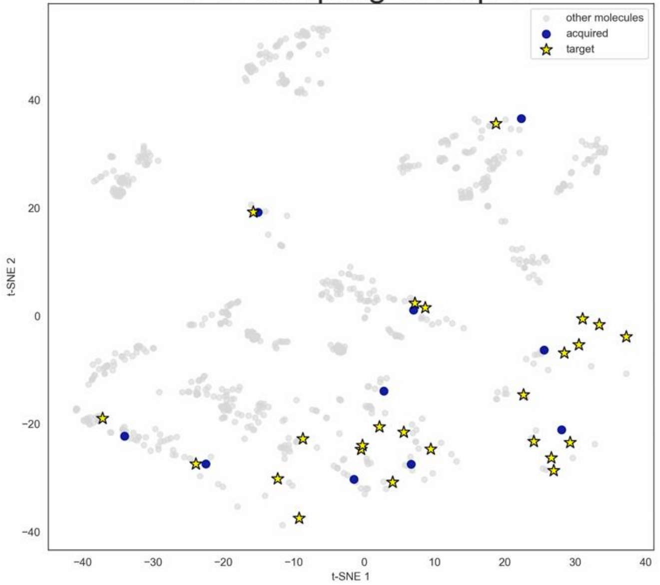
What makes a training molecule "good" for prediction on a target?



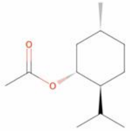
Active learning: leveraging uncertainty and predictions



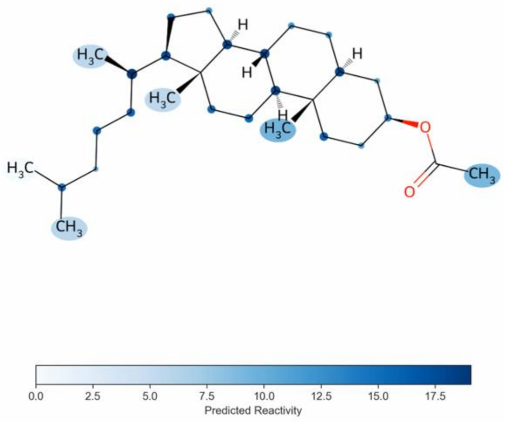
C–H sampling at step 1



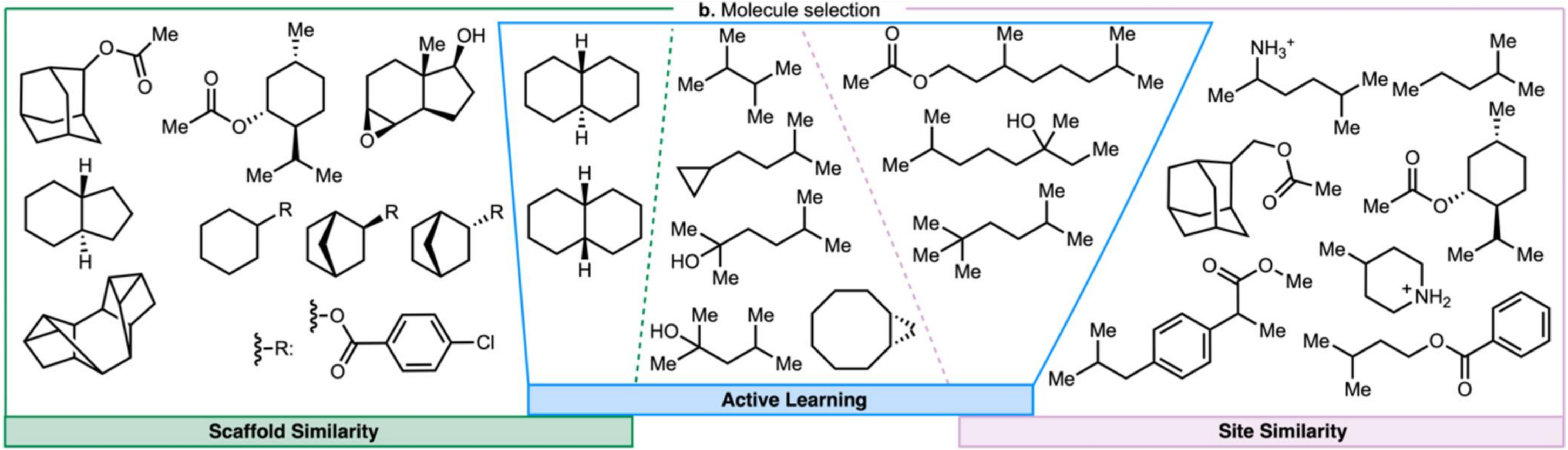
Tested molecules at step 1



Model prediction at step 1

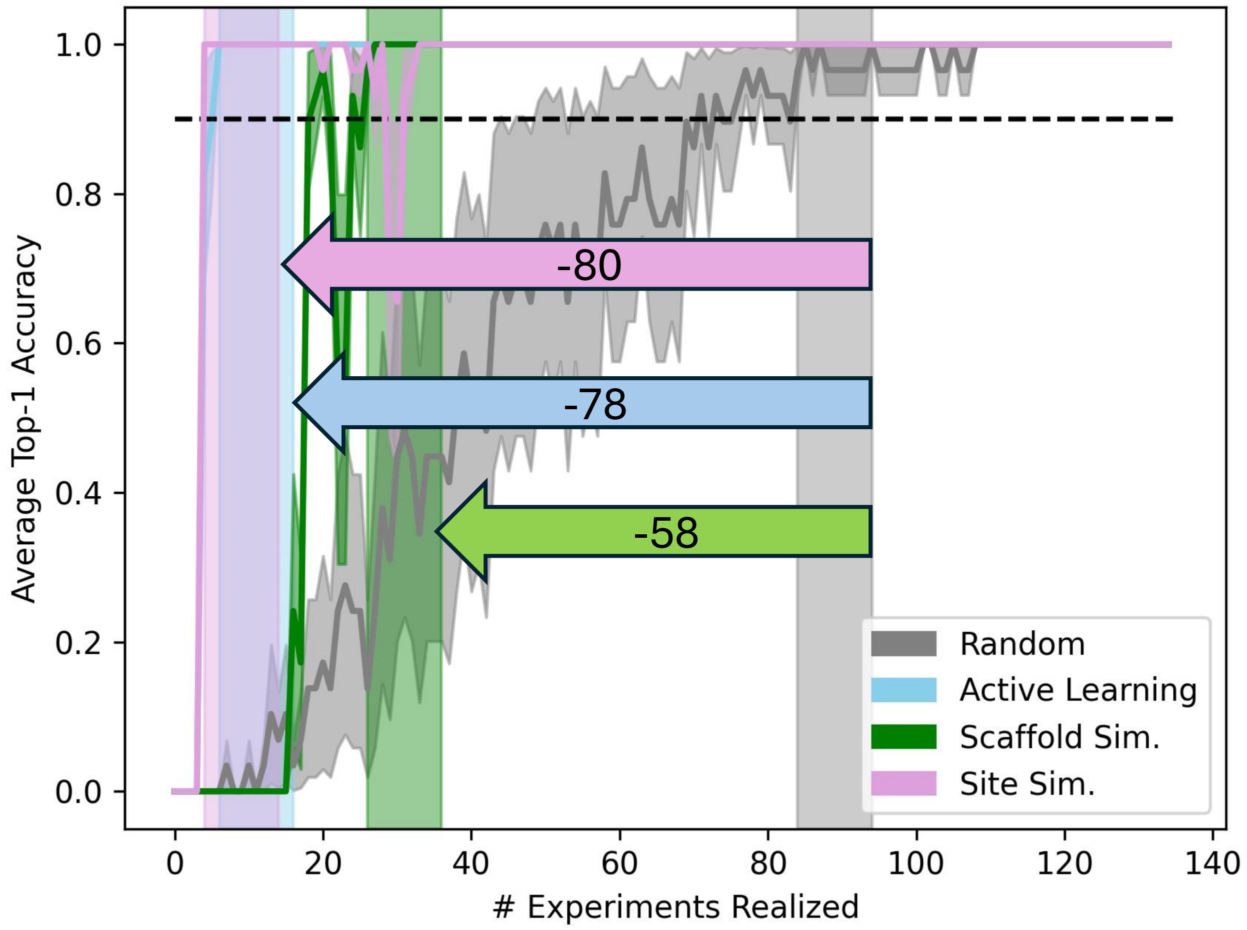


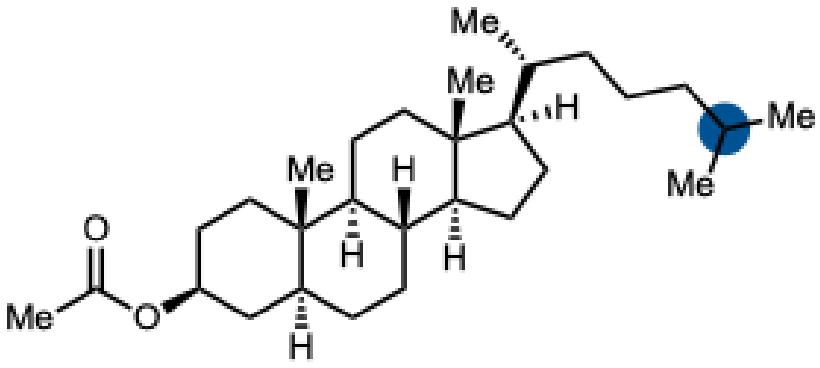
Analyzing Acquisition Function Selections



Designing target-specific datasets with acquisition functions

Learning curves:

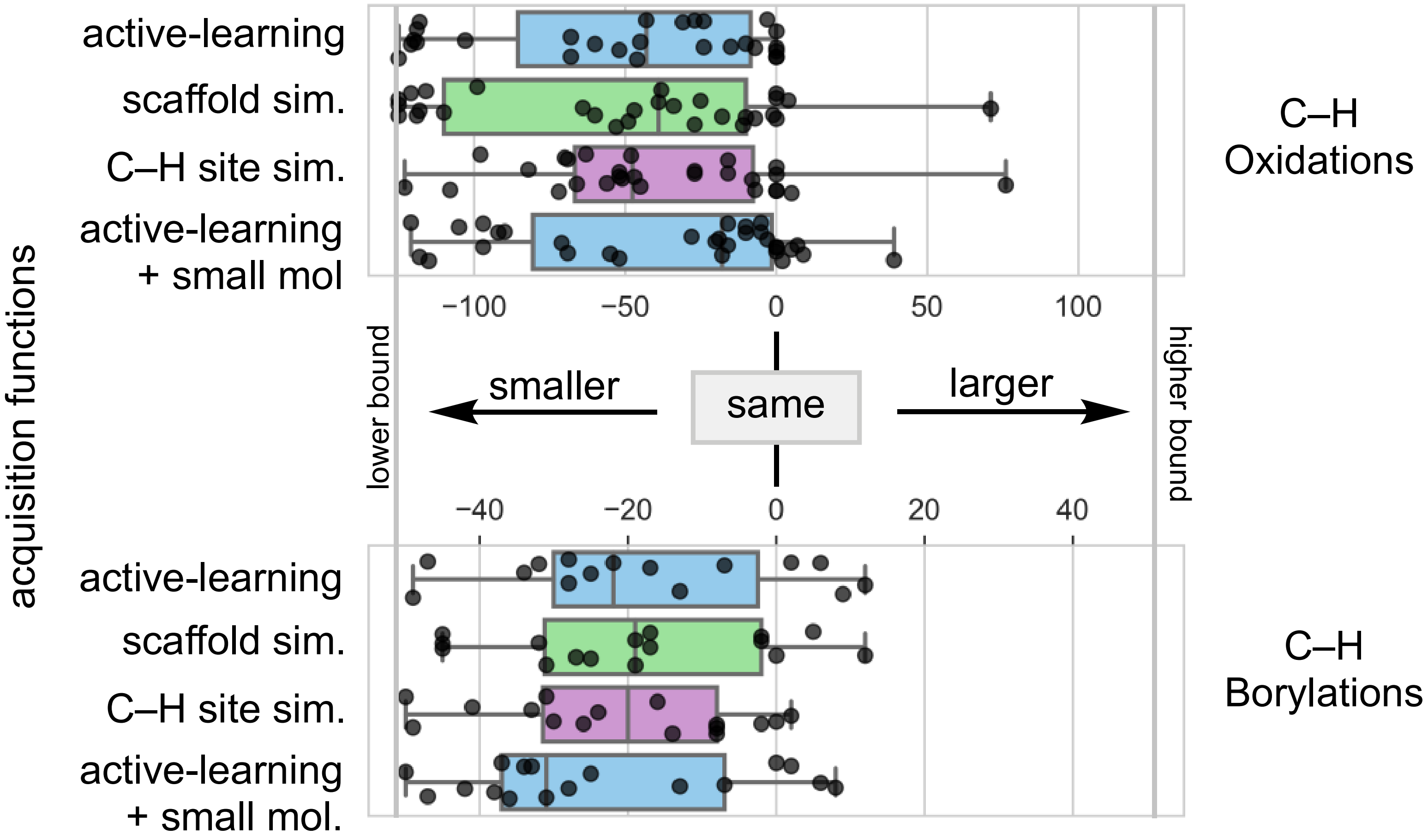




ground truth

Statistical significance

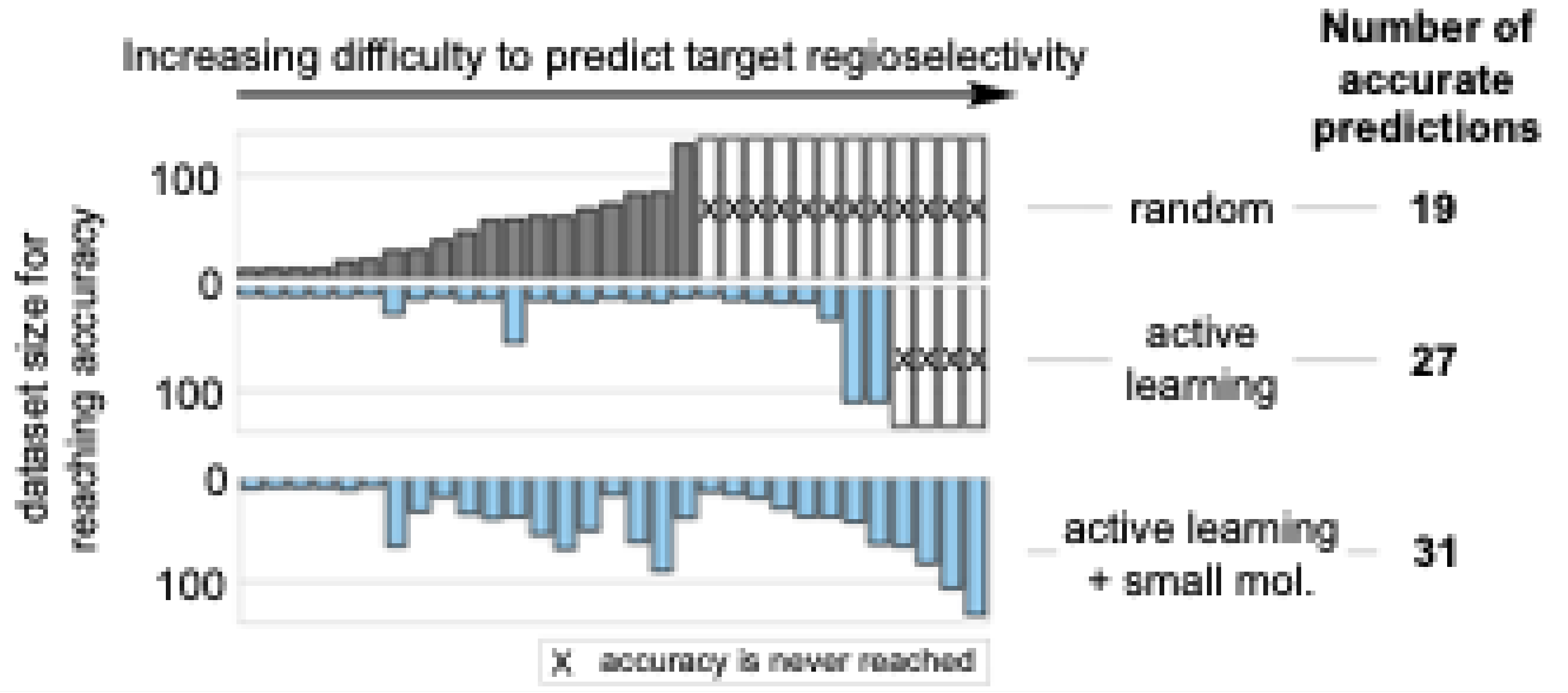
Dataset size difference between AFs and random



- Good performance on 5 complex molecules tested experimentally
- Good performance for predicting radical arene borylation using another literature dataset

Improvement of top-1 accuracy

Comparison of random and active learning AFs



More data is not always better: smaller datasets that cover only the relevant region of chemical space outperform larger datasets

Designing target-specific datasets with acquisition functions

STATISTICALLY SIGNIFICANT:

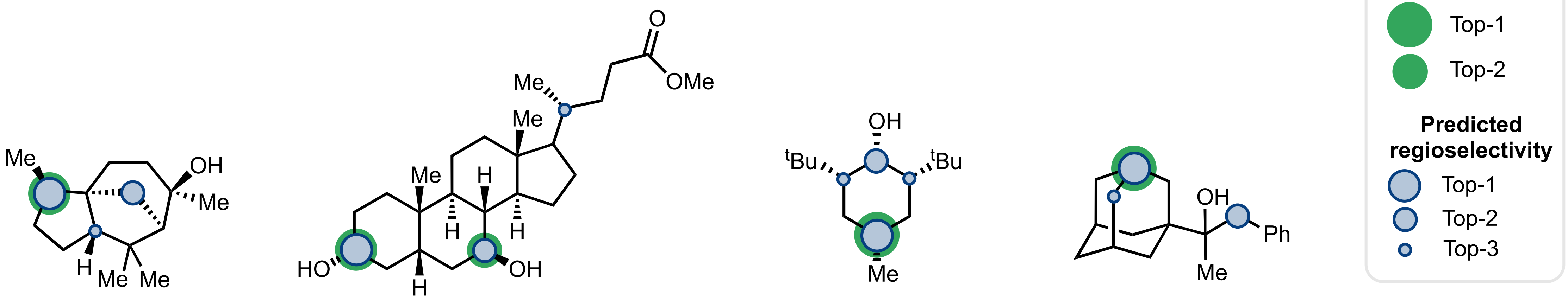
- Active Learning reduces ~40 datapoints
- Active Learning design specific datasets successful where random can't

TRANSPOSABLE TO DIFFERENT C–H FUNCTIONALIZATION:

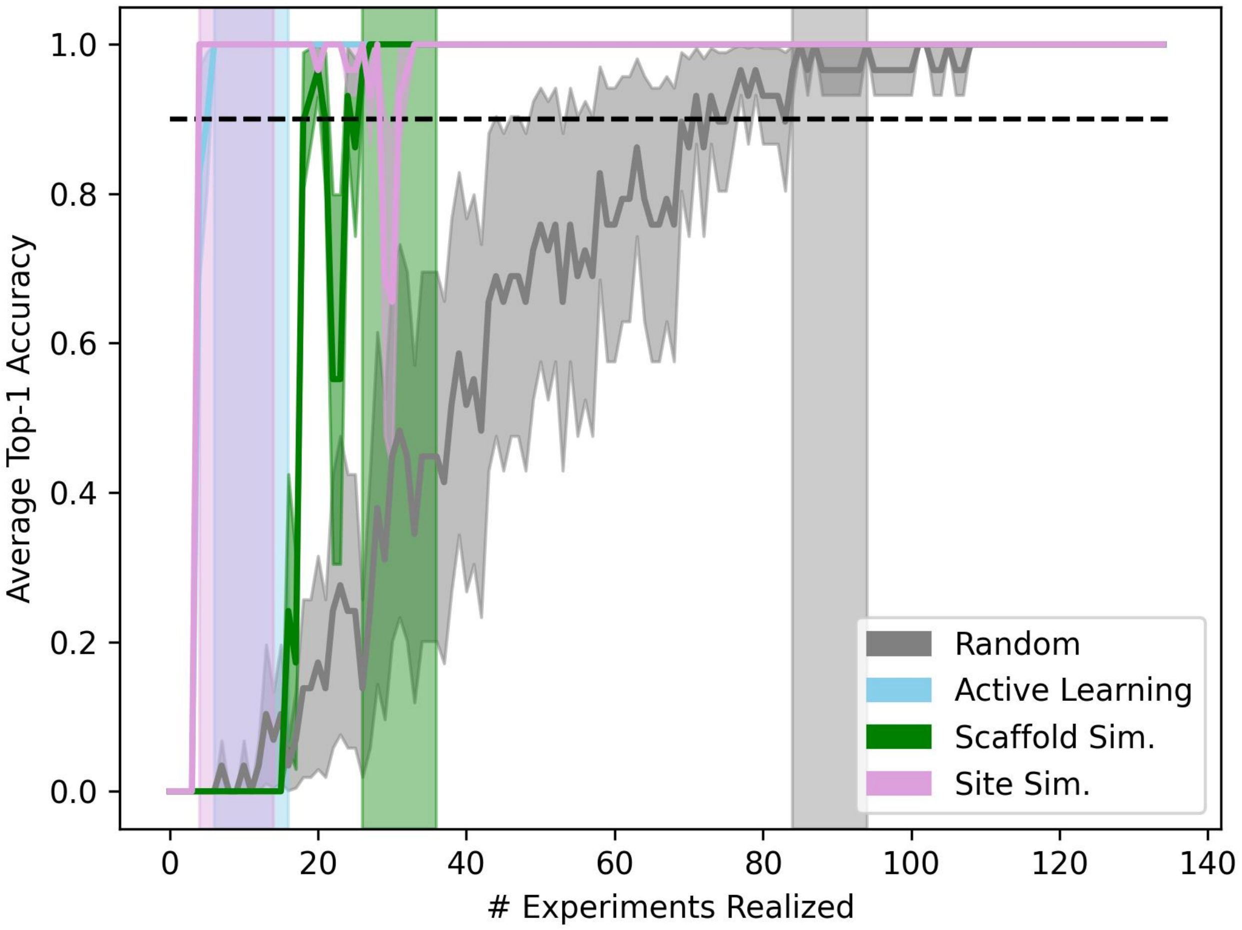
- $C(sp^3)$–H Dioxirane oxidations
- $C(sp^2)$–H borylations

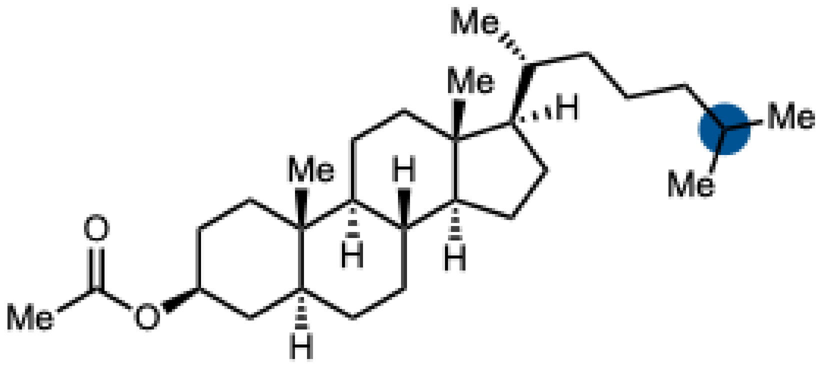
EXPERIMENTAL VALIDATION:

5 targets successfully predicted with this method



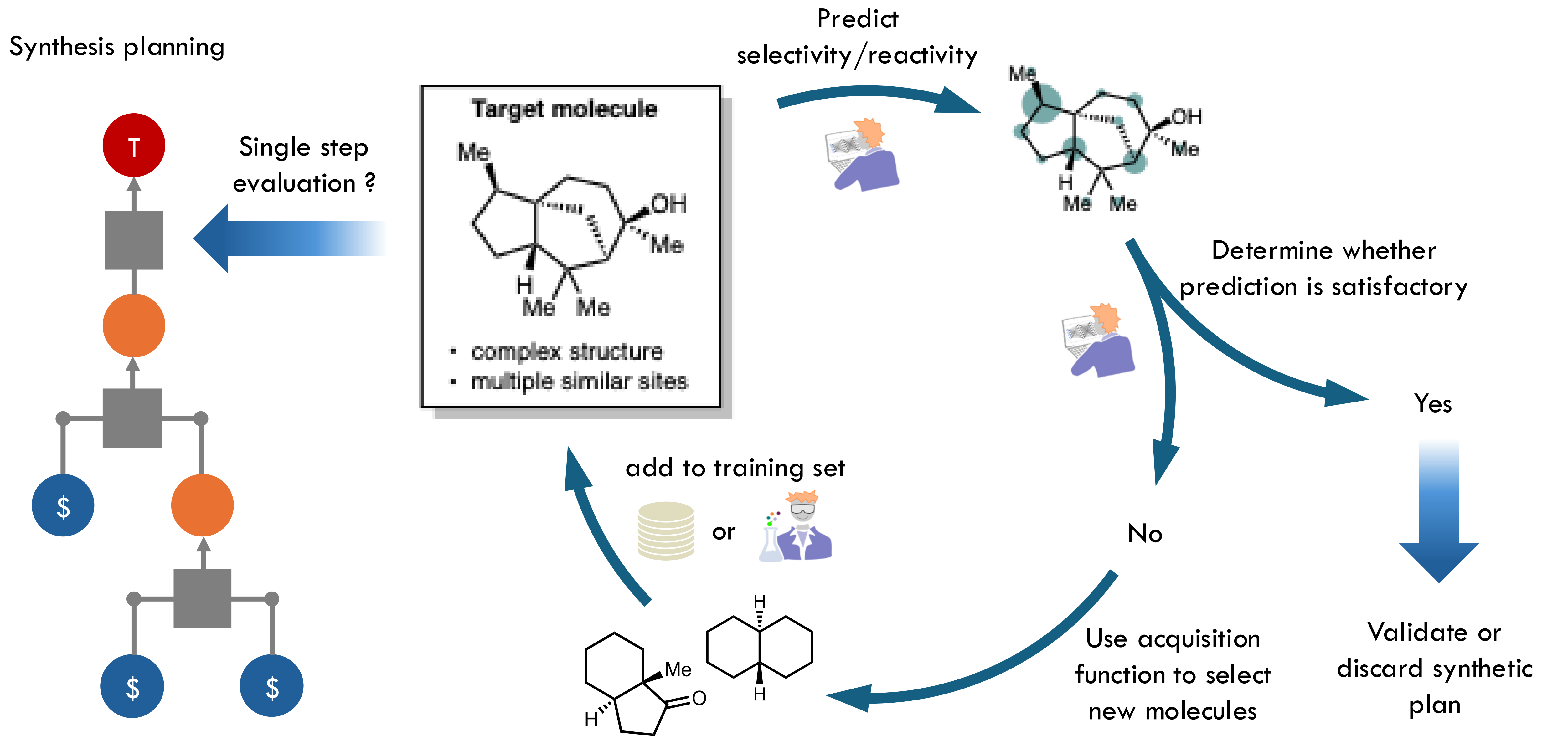
Future work: prospective use needs stopping criterion





ground truth

Perspective for de-risking synthetic planning

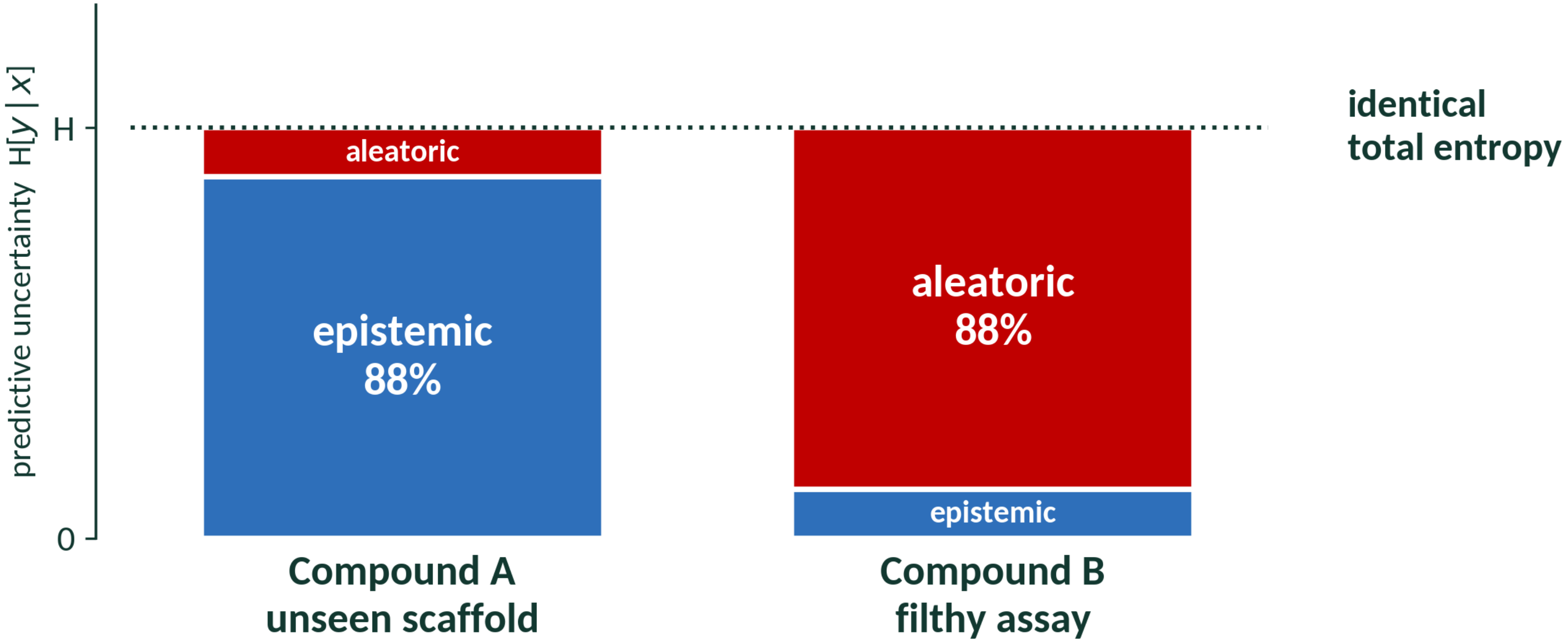


Plan

What are the sources of uncertainty?

Compound A
A scaffold the model has never seen. Wide error bars.

Compound B
A well-understood scaffold, but the assay is filthy. Replicates scatter by 30%.



Not all uncertainty is worth sampling

$$\mathbf{H}[\, y \mid x, \mathcal{D}\,] \quad = \quad \mathbb{I}[\, y \,;\, \omega \mid x, \mathcal{D}\,] \quad + \quad \mathbb{E}_{\omega}\left[\, \mathbf{H}[\, y \mid x, \omega\,]\,\right]$$

TOTAL

"I don't know."

EPISTEMIC

"I don't know YET — an experiment fixes this."

Sample x!

ALEATORIC

"Nobody knows. Running it again will not help."

Don't sample x!

Identity: Houlsby, Huszár, Ghahramani & Lengyel, arXiv:1112.5745, 2011 · vocabulary: Hüllermeier & Waegeman, Mach. Learn. 2021, 110, 457

Bayesian Active Learning by Disagreement (BALD)

$\mu(x)$ $\sigma^2(x)$

$\mu(x)$ $\sigma^2(x)$

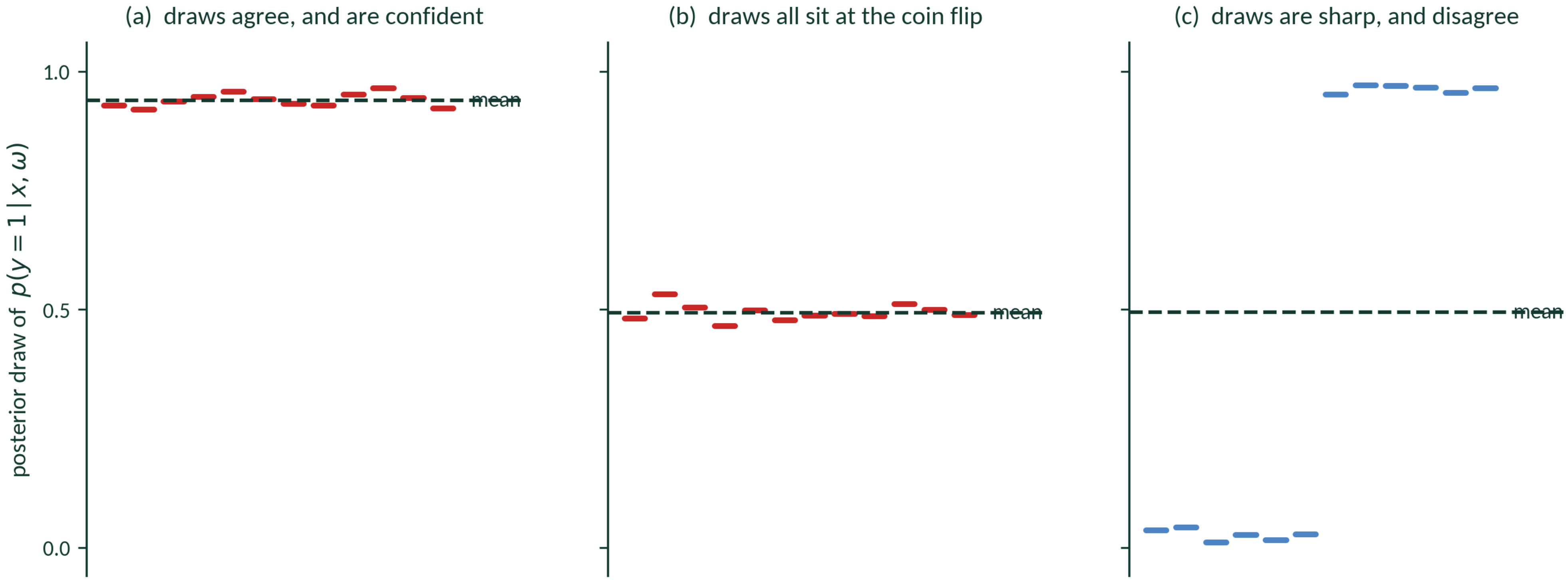
$\mu(x)$ $\sigma^2(x)$

$\mu(x)$ $\sigma^2(x)$

1. *Average $\sigma^2(x)$ is high and Standard Deviation of $\mu(x)$ is low: aleatoric uncertainty*

2. *Average $\sigma^2(x)$ is low and Standard Deviation of $\mu(x)$ is high: epistemic uncertainty*

Example on classification



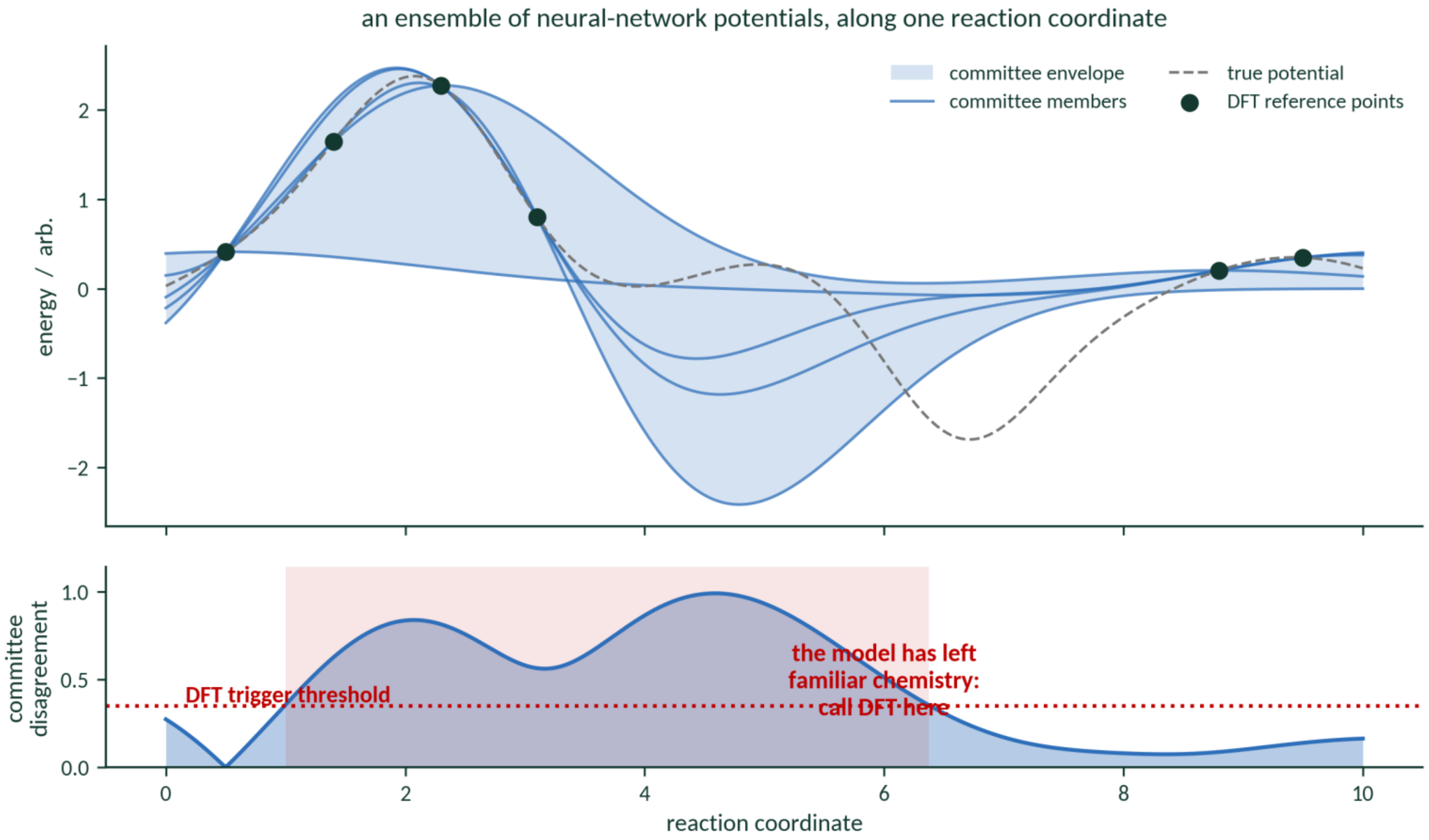
H = 0.33 bits BALD = 0.00 bits

H = 1.00 bits BALD = 0.00 bits

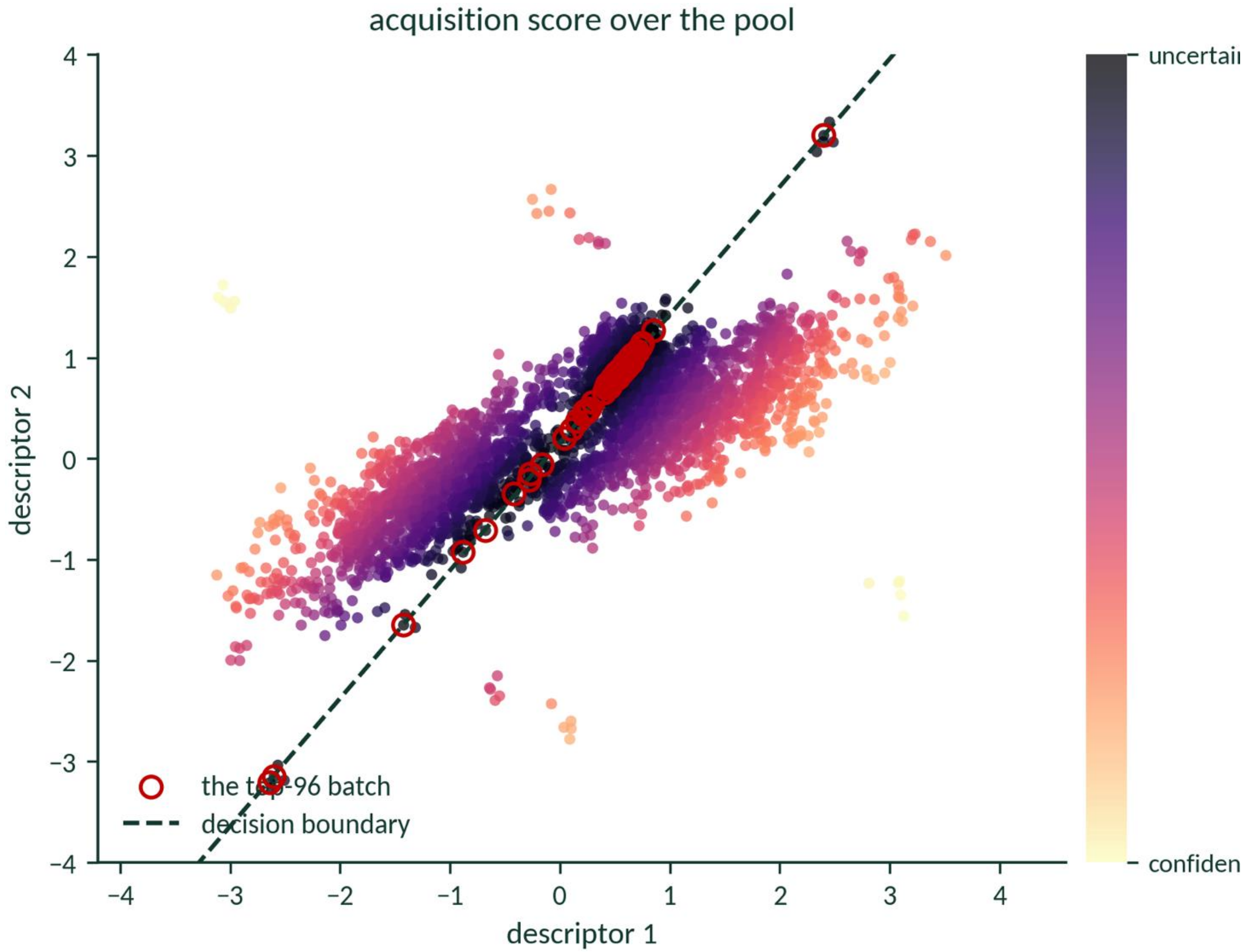
H = 1.00 bits BALD = 0.80 bits

BALD — Houlsby et al., arXiv:1112.5745, 2011 · H and BALD computed on the figure

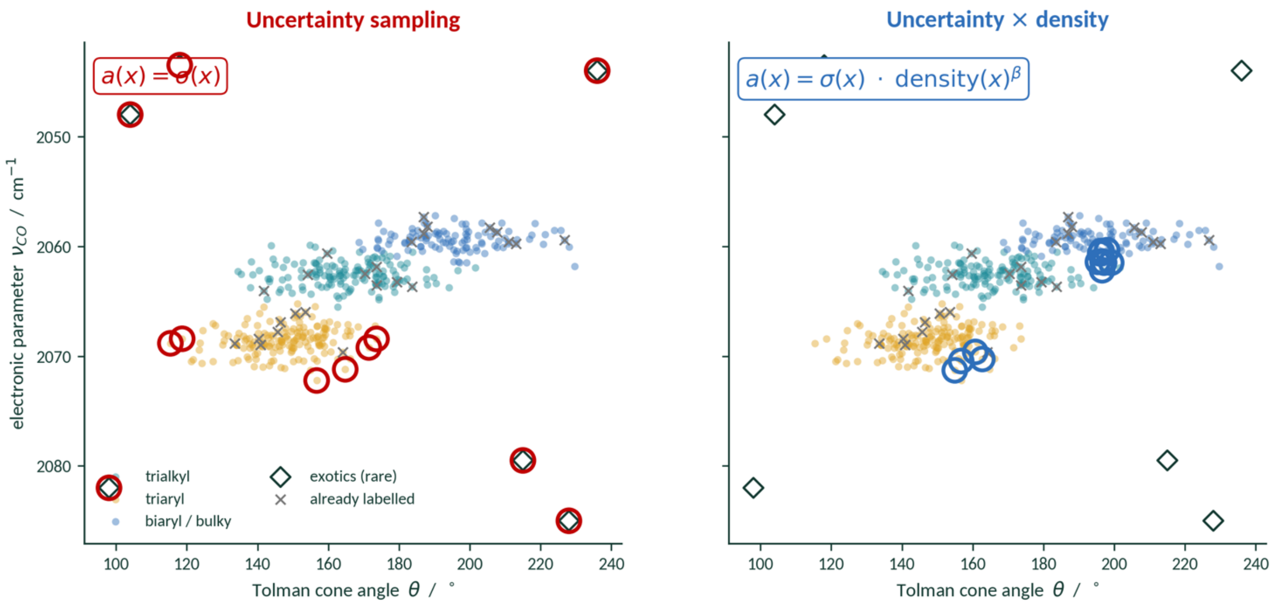
In practice: query by committee



Batching


acquisition score over the pool
descriptor 1
descriptor 2
the top-96 batch
decision boundary
uncertai
confiden
−4
−3
−2
−1
0
1
2
3
4

Plan



Weight by density

$\sigma(x) \cdot \text{density}(x)^\beta$ — reward points that are typical as well as uncertain

Constrain the pool

filter for synthesizability, cost, characterisability BEFORE scoring

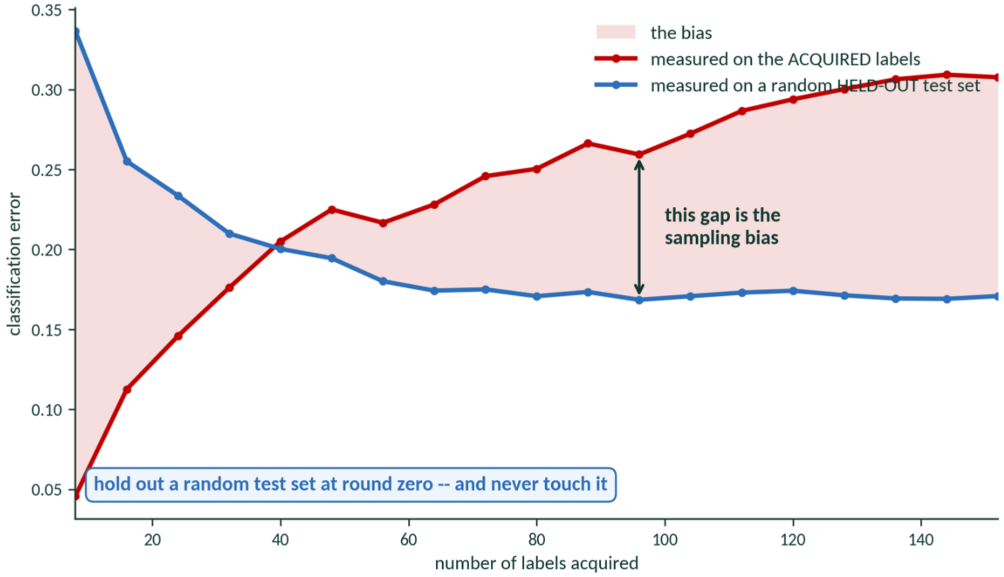
Bias deliberately

AF_AL toward small molecules took 27 targets to 31

Constrain the pool, not just the score.

2 · Your data is now a biased sample

Error estimation assumes $x_i \sim p(x)$, i.i.d. Under active learning $x_i \sim q_i(x)$ — an adaptive, model-dependent proposal. Your training set over-samples hard points, so it over-states difficulty.



uncertainty sampling, illustrative simulation, 30 runs

Hold out a random test set at round zero.

Never touch it. Never evaluate on the actively-acquired pool.

The successor-model problem.

Data selected by model A can be WORSE than random for model B. If you will ever swap your surrogate — and in molecular ML you always do — an AL-curated data set is a liability.

The bias is not always bad.

For overparameterised models it acts like hard-example mining. De-bias your EVALUATION always; de-bias your TRAINING only sometimes.

Lowell, Lipton & Wallace, EMNLP-IJCNLP 2019 · Farquhar, Gal & Rainforth, ICLR 2021, arXiv:2101.11665

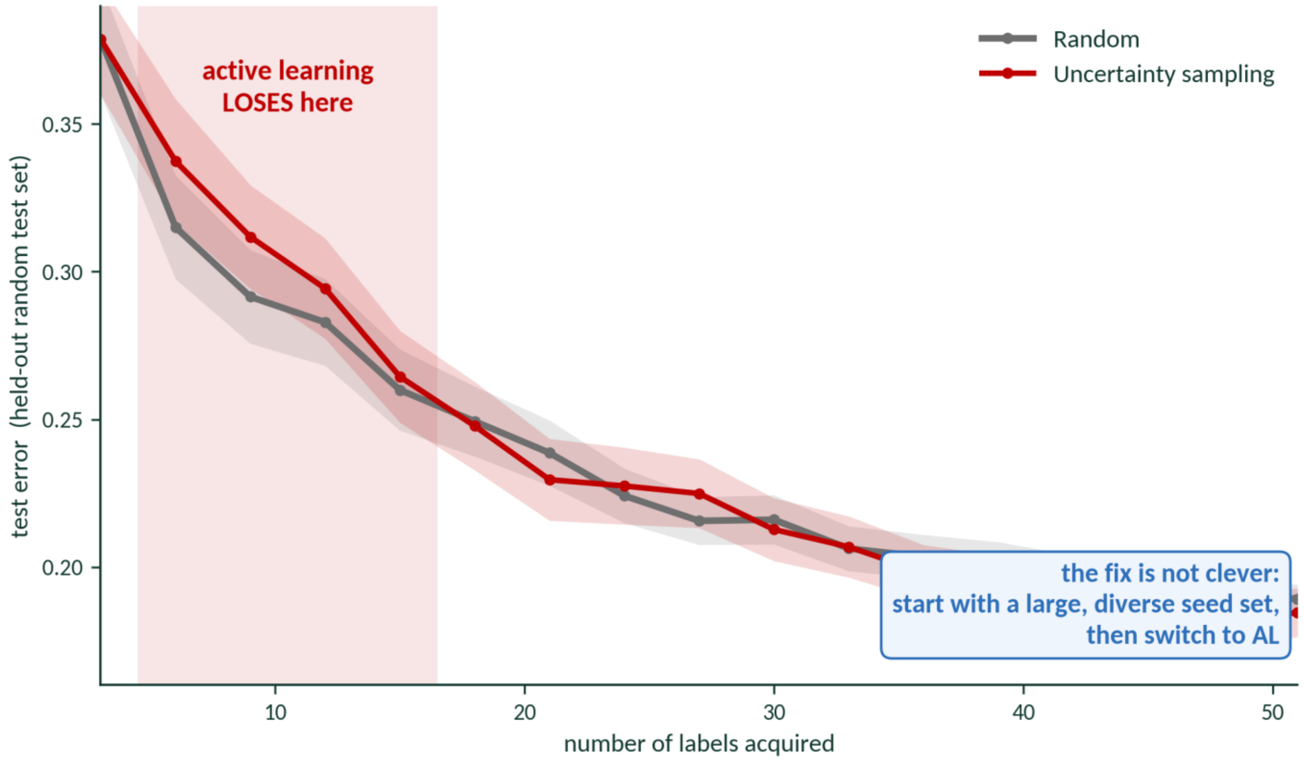
3 · When do you stop?

Criterion	The rule	Where it breaks
Budget / economic	stop when the money or the plates run out	honest, and what actually happens — but it is not a criterion
Max uncertainty	stop when max $a(x)$ over the pool falls below ε	overconfident models trip it far too early
Learning-curve plateau	stop when the validation metric flattens	INVALID if the validation set was actively acquired — see the last slide
Residual fail rate	stop when $<$ 5% of the remaining pool still fails your error	needs a tolerance in physical units, fixed in advance

Smith et al. used the last one, and it worked — because they wrote down 0.04 kcal/mol before they started.

So the question to take away: what is your 0.04 kcal/mol?

4 · And random is a much better baseline than you expect



illustrative simulation, 40 runs each; seed set of 3, batches of 3

Worse than random at n = 6, 9, 12, 15. Crosses over at 18.

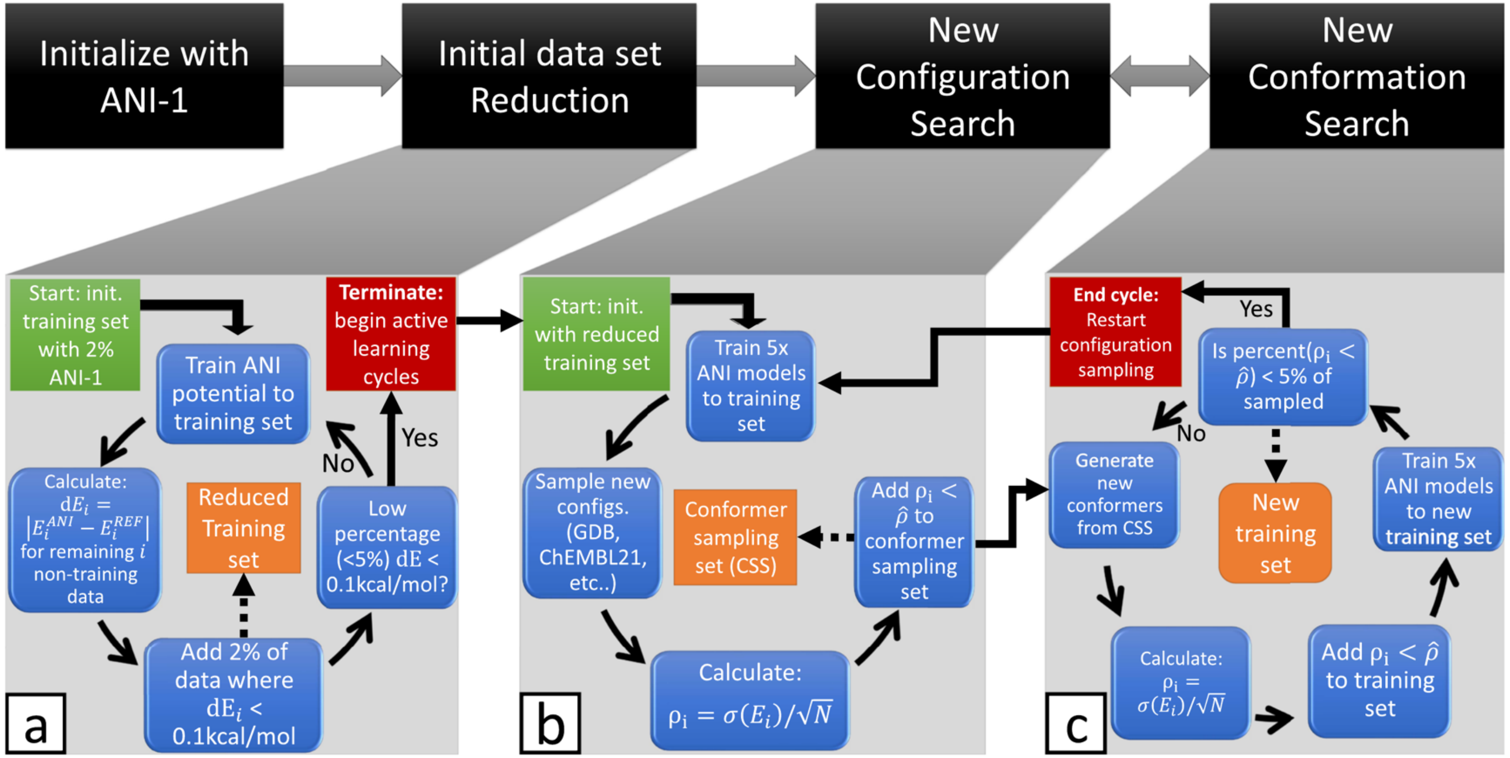
A model trained on almost nothing produces biased queries.
It does not yet know enough to know what it does not know.

Always plot the random baseline on the same axes.
In batch mode, Sener & Savarese found that oracle-quality uncertainty still lost to random.

A large fraction of published active-learning gains disappear the moment you draw that line.

Plan

"Less is more" — active learning for a universal potential



$$\rho_i = \sigma(E_i) / \sqrt{N} \qquad \text{add if } \rho_i > \hat{\rho}$$

$\hat{\rho}$ = 0.23 kcal/mol — 98% of errors above 1.5 kcal/mol

beats random-sampled ANI-1 with

10%

of the data

and vastly beats it at

25%

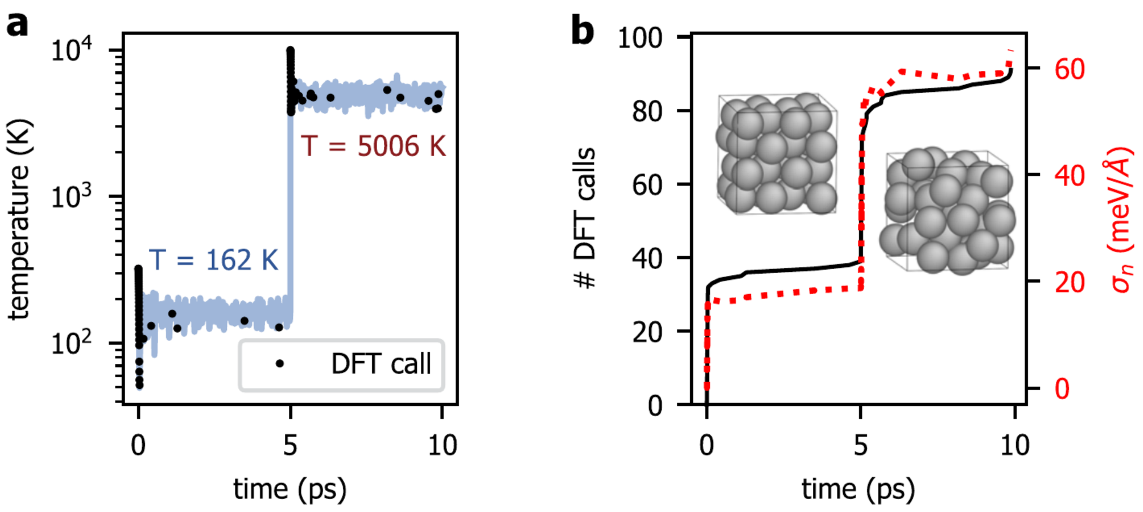
stops below

5%

fail cases remaining

Smith, Nebgen, Lubbers, Isayev & Roitberg, J. Chem. Phys. 2018, 148, 241733 · DOI 10.1063/1.5023802 · Fig. 2, reproduced for non-commercial teaching use

On the fly: the model asks for DFT when it gets lost



Candidates	the configurations the MD is visiting right now — they arrive, they are not enumerated
Oracle	DFT
Trigger	sparse-GP epistemic variance on the forces exceeds the model's own noise level
Bought	no further DFT after ~400 ps of vacancy diffusion; >300× faster than AIMD

Aluminium, 10 ps. It melts at t = 5 ps — and the DFT calls and the fitted noise both jump at exactly that moment.

Vandermause et al., npj Comput. Mater. 2020, 6, 20 · DOI 10.1038/s41524-020-0283-z · Fig. 3a,b, CC BY 4.0

Most of what is called active learning is optimization

Three famous, excellent papers. All three use the machinery from this lecture. None of them wants a globally good model.

Molecular pool-based AL

MolPAL

100 M docking candidates, D-MPNN surrogate, UCB or greedy. Recovers 94.8% of the top-50,000 after docking 2.4% of the library.

It wants the TOP OF THE RANKING. Error in the boring 97% is irrelevant.

Graff, Shakhnovich & Coley, Chem. Sci. 2021, 12, 7866

Autonomous synthesis

A-Lab

Recipes proposed from the literature, robotic synthesis, automated XRD, recipes updated after failures. 41 compounds from 58 targets in 17 days.

It wants NEW COMPOUNDS. A model of everything it did not make is worthless to it.

Szymanski et al., Nature 2023, 624, 86 · see Author Correction, Nature 2026, 650, E1

Pareto active learning

PAL

Uses the Pareto-dominance relation directly, so no scalarisation and no imposed weighting. Recovers the front with >98% fewer evaluations than random.

Its own first sentence: “the holy grail of material science is to find the OPTIMAL material.”

Jablonka et al., Nat. Commun. 2021, 12, 2312

Plan

1 · The BO/AL loop

2 · Objective = model improvement

3 · What changes in the acquisition function

4 · A worked example: regioselectivity

5 · Which uncertainty is worth buying?

6 · Practical issues

7 · Active learning in the wild — and what is not

8 · Conclusion

What to take away

The objective decides the acquisition function.

A value across a library → uncertainty, weighted by representativeness. A class or a ranking → margin. Ask what you are scoring before you ask which formula.

Not all uncertainty is worth sampling.

Epistemic shrinks when you spend money. Aleatoric does not. If your noise model does not depend on x, you cannot tell them apart.

Measure yourself honestly.

A random test set held out at round zero, a random baseline on the same axes, and an error tolerance written down before you start.

References — theory and method

★ Smith, J. S.; Nebgen, B.; Lubbers, N.; Isayev, O.; Roitberg, A. E. Less is more: sampling chemical space with active learning. J. Chem. Phys. 2018, 148, 241733. DOI 10.1063/1.5023802

★ Schleinitz, J.; Carretero-Cerdán, A.; Gurajapu, A.; et al. Designing Target-specific Data Sets for Regioselectivity Predictions on Complex Substrates. J. Am. Chem. Soc. 2025, 147, 7476–7484. DOI 10.1021/jacs.4c15902

Settles, B. Active Learning Literature Survey. Comp. Sci. Tech. Report 1648, Univ. Wisconsin–Madison, 2009. minds.wisconsin.edu/handle/1793/60660

Houlsby, N.; Huszár, F.; Ghahramani, Z.; Lengyel, M. Bayesian Active Learning for Classification and Preference Learning. arXiv:1112.5745, 2011.

Hüllermeier, E.; Waegeman, W. Aleatoric and epistemic uncertainty in machine learning. Mach. Learn. 2021, 110, 457–506. DOI 10.1007/s10994-021-05946-3

Seung, H. S.; Opper, M.; Sompolinsky, H. Query by committee. COLT '92, 287–294. DOI 10.1145/130385.130417

Kirsch, A.; van Amersfoort, J.; Gal, Y. BatchBALD. NeurIPS 32, 2019. arXiv:1906.08158

Sener, O.; Savarese, S. Active Learning for CNNs: A Core-Set Approach. ICLR 2018. arXiv:1708.00489

Settles, B.; Craven, M. An analysis of active learning strategies for sequence labeling tasks. EMNLP 2008, 1070–1079. aclanthology.org/D08-1112

Baum, E. B.; Lang, K. J. Query learning can work poorly when a human oracle is used. IJCNN 1992, Beijing. (no DOI)

References — chemistry

Vandermause, J.; et al. On-the-fly active learning of interpretable Bayesian force fields for atomistic rare events. npj Comput. Mater. 2020, 6, 20. DOI 10.1038/s41524-020-0283-z

Nippa, D. F.; Atz, K.; Hohler, R.; et al. Enabling late-stage drug diversification by high-throughput experimentation with geometric deep learning. Nat. Chem. 2024, 16, 239–248. DOI 10.1038/s41557-023-01360-5

Graff, D. E.; Shakhnovich, E. I.; Coley, C. W. Accelerating high-throughput virtual screening through molecular pool-based active learning. Chem. Sci. 2021, 12, 7866. DOI 10.1039/D0SC06805E — optimization, not model-building (slide 43)

Jablonka, K. M.; et al. Bias free multiobjective active learning for materials design and discovery. Nat. Commun. 2021, 12, 2312. DOI 10.1038/s41467-021-22437-0 — Pareto-front optimization (slide 43)

Szymanski, N. J.; et al. An autonomous laboratory for the accelerated synthesis of inorganic materials. Nature 2023, 624, 86–91. DOI 10.1038/s41586-023-06734-w — see Author Correction: Nature 2026, 650, E1. DOI 10.1038/s41586-025-09992-y

Reker, D.; Schneider, G. Active-learning strategies in computer-assisted drug discovery. Drug Discov. Today 2015, 20, 458–465. DOI 10.1016/j.drudis.2014.12.004

Podryabinkin, E. V.; Shapeev, A. V. Active learning of linearly parametrized interatomic potentials. Comput. Mater. Sci. 2017, 140, 171–180. DOI 10.1016/j.commatsci.2017.08.031

Lowell, D.; Lipton, Z. C.; Wallace, B. C. Practical Obstacles to Deploying Active Learning. EMNLP-IJCNLP 2019, 21–30. aclanthology.org/D19-1003

Farquhar, S.; Gal, Y.; Rainforth, T. On Statistical Bias In Active Learning: How and When To Fix It. ICLR 2021. arXiv:2101.11665

Gorantla, R.; et al. Benchmarking Active Learning Protocols for Ligand-Binding Affinity Prediction. J. Chem. Inf. Model. 2024, 64, 1955–1965. DOI 10.1021/acs.jcim.4c00220